CIF Variable, Unit, and Equation Reference

A Variable and Unit Reference

This appendix catalogs the symbols, dimensionless quantities, and key equations of the dissertation's analytical, design, and validation framework, with SI units, the practical units used in the CIF lineage, and step-by-step dimensional derivations. It spans both performance axes, the filtrate fraction (yield) and the particle-size cutoff (selectivity), together with the verification-and-validation layer that connects the analytical framework to numerical (COMSOL) and published-device evidence. Direct quotations from the source papers anchoring each unit convention and each adjudicated physical criterion are collected in Section A.10; the selectivity axis is validated in Section A.7 and the yield axis in Section A.6; an independent cross-lineage dataset is given in Section A.11; the design envelope is inventoried in Section A.8; and a consolidated master cheat-sheet is in Section A.12.

A.1 Geometric Variables

Table 1 fixes the geometric symbols and the in-plane versus out-of-plane convention. G, Λ and ϕ are included here because the wall porosity is one of the three model gaps to which the cutoff residual is attributed (Section A.4).

Evidentiary markers. Where a row's status is not obvious it is marked EXTERNAL (an input the record does not derive: a source states it, or this work supplies it and no validator discharges it), DERIVED (it follows from geometry or from an Appendix A result), or *This work* (a convention this document sets). Unmarked rows are EXTERNAL and carry their locator in the Source column. The status columns of later tables use five more markers: FITTED (calibrated to data by a fit), NUMERICAL (computed by a numerical solve), MEASURED (set by a measurement), PROPOSED (a parameterization this work proposes and has not yet computed or measured) and BOUNDS (a condition that limits where the design relations hold); ESTABLISHES and COMPETES, in the name column of Table 4, say what a quantity does in the argument.

Table 1: Geometric variables used to specify CIF device architecture. Practical units reflect the CIF lineage convention; SI units are used in all dimensional derivations.

Symbol	Quantity	SI unit	Practical unit	Source
w_c	Center-channel width (constant-w_c regime)	m	µm	[1, p. 4498, Fig. 1]
$w_c(i)$	Center-channel width of segment i, after gap row i (tapered-w_c regime; $w_c(0)$ is the inlet width). Xia's width at gap i is read as this $w_c(i)$ in Sections A.7 and A.9, as in App. A; the width arriving at row i is $w_c(i-1)$, one taper step away	m	µm	[5, p. 916]; [2, p. 10 (Sci Rep PDF pagination)]
w_s, $w_s(i)$	Side-channel width (generic / of segment i, after gap row i, the width that sets $R_s(i)$)	m	µm	[1, p. 4498, Fig. 1]
w_c(ref)	Reference center-channel width (Xia 2022 scaling)	m	µm	[3, p. 1]
w_{ch}	*Generic* channel width placeholder, $w_{\mathrm{ch}} \in \{w_c, w_s\}$, used only inside the resistance law (Eqn 6)	m	µm	This work

Table 1 (continued)

Symbol	Quantity	SI unit	Practical unit	Source
w	*Diverted fluid lamina width* (empirical extracted-lamina width; cutoff reference length)	m	µm	[5, p. 917]
δ	Dividing-streamline (skim) width, $\delta = w_c\sqrt{f_{\text{gap}}/3}$ at leading order; plays the role of w in the cutoff law. DERIVED (App. A, Result 2)	m	µm	[9] (`delta_skim`); [13, p. 656]
b	First-streamline (boundary-lamina) width; Inglis writes $D_c = 2\beta$, rendered b here (see Lamina-width collision below)	m	µm	[13, p. 656, Eqn 1]
h	Channel depth (out-of-plane height); Gifford's d in the Eqn 5 sentence ("d is the depth of the channel")	m	µm	[1, p. 4498]
d	*[deprecated]* Gifford depth notation $\equiv h$; retained for cross-reference only. Use h to avoid collision with d_{crit}.	m	µm	[1, p. 4498]
L_{dev}	Device length (along flow direction). *Renamed from L*: Gifford's L is the *segment* length, not the device length	m	mm	This work
$L_c(i), L_s(i)$	Center/side channel *segment* length after gap row i (between rows i and $i+1$; segment 0 is the inlet stretch, segment N the outlet), the segment that carries $Q_c(i)$ and $Q_s(i)$. Gifford describes a segment as lying between rows $i-1$ and i, one row earlier, which is immaterial for his constant f_{gap}; the quantity Gifford's Eqn 5 calls L, clarified there as "the size (along the direction of flow) of an obstacle in the device"	m	µm	This work, after Gifford's Eqn 5 L ([1, p. 4498]; clarified [1, p. 4499])
g	*[deprecated]* An earlier draft used g for the gap opening. That is the same quantity as G below, and lower-case g is needed for the cubic kernel; use G.	m	µm	—
G	**Link width**: the along-flow width of the opening in the wall between center and side channel. This is the dimension a particle passes through, and the one Gifford reports as "wide (∼20 µm) gaps"	m	µm	This work; value [1, p. 4498]
ℓ_g	**Link length**: the wall thickness the fluid crosses, i.e. the flow path *through* the opening. With G it sets the gap cross-section, and hence the gap resistance R_{gap} of Table 2; the hydraulic diameter of the opening is $D_h = 2Gh/(G+h)$	m	µm	This work
g_p	**Post width**: the along-flow width of the wall segment between successive openings. G and g_p are both along-flow, which is why they sum to a row pitch	m	µm	This work
Λ	Row pitch, $\Lambda(i) \geq \max(G + g_p,\ 2p$ [if folded]); equality only for an unfolded tight-packed row. A *design constraint*, not a definition: the bound is set by the minimum feature and fold geometry	m	µm	This work
ϕ	**Wall porosity**, $\phi = G/\Lambda$: the open fraction of the "wall" the resistance kernels assume solid. DERIVED from geometry. One of the *three* model gaps the cutoff residual is measured against (Section A.4). Distinct from ϕ_{cal}, which is EXTERNAL (Section A.8), and from the fold factor $\Phi(i)$	—	—	This work; [9]

Table 1 (continued)

Symbol	Quantity	SI unit	Practical unit	Source
N	Total number of gap rows. Gifford's symbol is i_T (Fig. 2 caption, "the necessary total number of rows of gaps, i_T"); N is this work's, adopted to free i for the row index	—	—	This work, after [1, p. 4499]
i	Gap row index, $1 \le i \le N$; flows and segments are labelled $0 \le i \le N$, 0 being the inlet stretch	—	—	[1, p. 4498, Fig. 1]

Convention. w_c, w_s (and the generic placeholder w_{ch}) are the in-plane channel widths (perpendicular to flow); h is the out-of-plane channel depth.

Lamina-width collision. The bare symbol w denotes the *diverted fluid lamina width* only. The rectangular-duct resistance law (Eqn 6) takes the *channel* width as argument; that argument is written w_{ch} here, never bare w. The theoretical lamina width is $\delta = w_c\sqrt{f_{\mathrm{gap}}/3}$; the empirical lamina width is w; the Inglis first-streamline width is b. Inglis writes this as $D_c = 2\beta$ with Greek β ([13, p. 656, Eqn 1]); it is rendered b here to free β for the cutoff scaling exponent. These three denote the same boundary lamina, and their identification $b \equiv \delta$ is a *proven* result, not a definition: the CIF skim cubic is algebraically identical to the Inglis DLD cubic (App. A, Result 4; recorded as a proven node in the argument map, App. B).

G/g collision: g is the cubic kernel, not a length. Lower-case g denotes the skim kernel $g(x) \equiv 3x^2 - 2x^3$ and rides as a subscript on the paired prefactor κ_g. It is dimensionless. It is *not* a gap width: the opening is G, its through-wall length is ℓ_g, and the post is g_p; G and g differ only by case and denote different kinds of thing, so a reader scanning a formula must read the case. An earlier draft of this table used g for the opening, which both duplicated G and collided with the kernel; that row is deprecated above. The gap geometry needs no lower-case g: $\Lambda(i) \ge \max(G + g_p,\, 2p)$ and $\phi = G/\Lambda$ are stated entirely in G, g_p and Λ.

Two further symbol collisions, declared. The lower-case ϕ above is the wall porosity; the upper-case $\Phi(i) = r(w_s(i))/r(\mathrm{GS})$ used in the folded-channel geometry is the *fold factor* and is a different quantity. The fold pitch $p = \mathrm{GS} + \max(F_{\min}, 2R_{\min})$ of that same geometry (GS the fold width, $F_{\min}$ the minimum feature and $R_{\min}$ the minimum inner-corner radius, as the design specification's symbol table defines them) shares a letter with the hydrostatic pressure p of Table 2; pressure is always written P or p in a pressure context and the fold pitch only inside that folded-channel geometry.

A.2 Fluid and Hydraulic Variables

Table 2 lists the flow, pressure and resistance variables, together with the two external inputs $\sigma(i)$ and σ_0. Two letters are overloaded and are disambiguated in notes below the table: R carries four unrelated meanings across this appendix, and P is both the pressure and the normalised march flow $P(i)$ within this one table.

Table 2: Fluid-mechanical and hydraulic variables.

Symbol	Quantity	SI unit	Practical unit	Source
Q_c, $Q_c(i)$	Center-channel volumetric flow rate after gap row i	$\mathrm{m^3\,s^{-1}}$	$\mu\mathrm{L\,min^{-1}}$	[1, p. 4498]
Q_s, $Q_s(i)$	Side-channel volumetric flow rate (one side, after gap i)	$\mathrm{m^3\,s^{-1}}$	$\mu\mathrm{L\,min^{-1}}$	[1, p. 4498]
$Q_{\mathrm{gap}}(i)$	Volumetric flow through one gap on one side at row i	$\mathrm{m^3\,s^{-1}}$	$\mathrm{nL\,min^{-1}}$	[5, p. 915]
Q_{in}	Total inlet flow rate. The lineage devices are operated at 30 $\mu\mathrm{L\,min^{-1}}$; the *symbol* is this work's	$\mathrm{m^3\,s^{-1}}$	$\mu\mathrm{L\,min^{-1}}$	This work; rate [5, p. 915]
$Q_m(i)$	Middle-channel flow rate at gap i (Dinh notation; $= Q_c(i-1)$)	$\mathrm{m^3\,s^{-1}}$	$\mu\mathrm{L\,min^{-1}}$	[5, p. 915]

Table 2 (continued)

Symbol	Quantity	SI unit	Practical unit	Source
P, p	Hydrostatic pressure. Gifford writes only the differential ΔP; the bare symbol is this work's	Pa	kPa or psi	This work
ΔP, $\Delta P(i)$	Pressure drop across one channel segment	Pa	kPa	[1, p. 4498]
$p_{\text{center}}(L_{\text{dev}})$	Pressure in center channel at the outlet	Pa	kPa	Condition [1, p. 4500]; symbol this work
$p_{\text{side}}(L_{\text{dev}})$	Pressure in side channel at the outlet	Pa	kPa	Condition [1, p. 4500]; symbol this work
R_c, $R_c(i)$	Hydraulic resistance of one center-channel segment	$\text{Pa}\,\text{s}\,\text{m}^{-3}$	—	[1, p. 4498]
R_s, $R_s(i)$	Hydraulic resistance of one side-channel segment	$\text{Pa}\,\text{s}\,\text{m}^{-3}$	—	[1, p. 4498]
R_{gap}	Hydraulic resistance of one gap. Gifford describes the gaps as presenting "a relatively small amount of resistance" but defines no symbol; this one is this work's	$\text{Pa}\,\text{s}\,\text{m}^{-3}$	—	This work, after [1, p. 4498]
$P(i)$, $S(i)$	Normalised center and side flows along the march, $P(i) = Q_c(i)/Q_c(0)$ and $S(i) = Q_s(i)/Q_c(0)$. *Not* the pressure P above: see the collision note	—	—	This work; [9]
$\sigma(i)$, σ_0	Cumulative side-flow ratio along the march, $\sigma(i) = S(i)/P(i)$; σ_0 is the (fictitious) inlet side-flow. EXTERNAL: enters the mass identity $P(i) + 2S(i) = 1 + 2\sigma_0$ and is not derived here	—	—	This work; [9]
μ	Dynamic viscosity of fluid	$\text{Pa}\,\text{s}$	$\text{mPa}\,\text{s}$	[1, p. 4498]
ρ	Fluid density. Water is taken as 1000 $\text{kg}\,\text{m}^{-3}$ and platelet-rich plasma as 1025 $\text{kg}\,\text{m}^{-3}$; neither value is measured here	$\text{kg}\,\text{m}^{-3}$	$\text{kg}\,\text{m}^{-3}$	This work (assumed)
$u(y)$, v	Local fluid velocity	$\text{m}\,\text{s}^{-1}$	$\text{mm}\,\text{s}^{-1}$	Standard
$u_{\max}$	Peak velocity in parabolic profile	$\text{m}\,\text{s}^{-1}$	$\text{mm}\,\text{s}^{-1}$	Standard

R-symbol collision. Four distinct objects use the letter R in this dissertation: the hydraulic resistances R_c, R_s, R_{gap} (this table); the depth-correction *function* $R(\alpha)$ (Section A.4); the Reynolds number Re (Table 3); and the multi-variant ratio $\mathcal{R}$ (Eqn 17). To keep them separate, the depth correction is written K_{depth} throughout, with $R(\alpha)$ used only as its interpretive alias when the aspect-ratio dependence is being emphasized; $\mathcal{R}$ is always script.

P-symbol collision. The pressure P of this table and the normalised center flow $P(i)$ are different quantities that share a letter within this same table. Pressure is always written with a Δ or a subscript (ΔP, p_{center}, p_{side}) or as the bare P with no argument; the march quantity is *always* written with its row index, $P(i)$, and is paired with $S(i)$. The terminal filtrate fraction is $F = 1 - P(N)$, never $1 - P$.

Outlet pressure equilibration (P1.4). The two outlet pressures $p_{\text{center}}(L_{\text{dev}})$ and $p_{\text{side}}(L_{\text{dev}})$ are taken equal. This is not an assumption introduced here: the lineage source states the resistance-ratio calculation holds "provided the hydrostatic pressures in the output collection areas are kept equal" ([1, p. 4500]). It is the premise of the closed-form filtrate identity (Eqn 11), stated as P1.4 in the formal argument of Section A.10; App. B's C4 carries it as premise (3) of P-C4.1.

Unit equivalences.

$$1\ \mu\mathrm{L\,min^{-1}} = 1.667 \times 10^{-11}\ \mathrm{m^3\,s^{-1}}$$
$$1\ \mathrm{nL\,min^{-1}} = 1.667 \times 10^{-14}\ \mathrm{m^3\,s^{-1}}$$
$$1\ \mathrm{kPa} = 1000\ \mathrm{Pa} = 0.145\ \mathrm{psi}$$
$$1\ \mathrm{psi} = 6894.76\ \mathrm{Pa}$$
$$1\ \mathrm{mPa\,s} = 10^{-3}\ \mathrm{Pa\,s}$$

A.3 Dimensionless Quantities

Table 3 collects the dimensionless groups. Two entries are easily misread: f_{gap} is defined *per side and per gap*, which is the origin of the factor of two in the continuity relation, and F has two distinct forms with different explanatory roles (Section A.9). Three symbol collisions are declared in notes below the table.

Table 3: Dimensionless ratios and design parameters. All entries are strictly dimensionless; SI and practical columns are omitted.

Symbol	Name	Defining relation	Source
f_{gap}, $f_{\mathrm{gap}}(i)$	Per-gap filtration fraction (per-side, per-gap)	$Q_{\mathrm{gap}}(i) = f_{\mathrm{gap}}(i) \cdot Q_c(i-1)$	[5, p. 915], [1, p. 4498]
$f_{\mathrm{gap}}(\mathrm{ref})$	Reference f_{gap} in the Xia 2022 corrected scaling	Xia 2022 corrected Eqn 1	[3, p. 1]
f^*_{gap}	Empirical threshold f_{gap} for cutoff (read visually)	"the filtration fraction per gap below which the beads of a specific diameter are consistently" retained	[1, p. 4500]
F_{outlet}	Filtrate fraction from the outlet divider	$F_{\mathrm{outlet}} = 2R_c/(R_s + 2R_c)$; *contains no* f_{gap}	Eqn 11; this work, under the outlet condition of [1, p. 4500]
$F_{\mathrm{recursion}}$	Filtrate fraction from the per-gap march	$F_{\mathrm{recursion}} = 1 - \prod_i (1 - 2f_{\mathrm{gap}}(i))$	Eqn 12; derived from Eqn 3 of [1, p. 4498]
F_{V1}, F_{V3}	Filtrate fraction at the Dinh V1 and V3 variants	Closed form, paper, or COMSOL	[5, p. 919]
σ_F	One-SD experimental uncertainty in F. *Not* the side-flow ratio $\sigma(i)$ of Table 2: see the collision note	Propagated from measurement	[5, p. 919]
$\alpha = h/w_{\mathrm{ch}}$	Channel aspect ratio	—	This work; the resistance series it enters is [6, p. 875]
$\widetilde{d}$	Reduced size ratio, $\widetilde{d} = d_{\mathrm{crit}}/w_c$	Sets the plateau ($\widetilde{d} \geq 0.035$) and foot ($\widetilde{d} < 0.012$) bins	[9] (`kappa_bins`)
x	Reduced skim width, $x = \delta/w_c$	root of $f_{\mathrm{gap}} = 3x^2 - 2x^3$	[9] (`skim_invert`); [13, p. 656]
β	Cutoff f_{gap}-scaling exponent	$\beta = \partial \ln d_{\mathrm{crit}}/\partial \ln f_{\mathrm{gap}} = \frac{1}{2}$ (empirical plateau fit ≈ 0.51)	[9] (`exponents`); App. A, Result 6
γ	Cutoff w_c-scaling exponent	$\gamma = \partial \ln d_{\mathrm{crit}}/\partial \ln w_c = 1$ (empirical plateau fit ≈ 1.02)	[9] (`exponents`); App. A, Result 6
Re	Reynolds number	$\mathrm{Re} = \rho U L_{\mathrm{char}}/\mu$	[10], which defines "Re is the Reynolds number"

Table 3 (continued)

Symbol	Name	Defining relation	Source
ε	*Source notation, not a dissertation variable.* In the deterministic-lateral-displacement literature ε is the row-shift fraction, and the cross-family comparison of Section A.11 quotes prefactors "across $\varepsilon = 0.01$–0.5" in that sense. It is never used as a CIF quantity here	quoted only	[13, 18]
Π	Cumulative product (in $F_{\text{recursion}}$)	$\Pi = \prod_i (1 - 2f_{\text{gap}}(i))$	Derived from Eqn 3 of [1, p. 4498]

Convention. f_{gap} is the *per-side, per-gap* extraction fraction. The total bilateral extraction at gap row i is $2f_{\text{gap}}(i)$, which is the origin of the factor of 2 in the continuity equation (Eqn 3 of [1, p. 4498]) and in the closed-form identity numerator $2R_c$.

Aspect-ratio symbol. This appendix writes the channel aspect ratio as $\alpha = h/w_{\text{ch}}$ and reserves β for the cutoff-law scaling exponent. This is the canonical dissertation convention and supersedes the alternatives. Note that the reference mathematics takes the aspect ratio as a parameter literally named `beta` (`c2(beta)`, `K_depth(beta)`, evaluated at 0.80/1.40/2.69). When reading code, `beta` is this appendix's α; when reading this appendix, β is never an aspect ratio.

σ-symbol collision. σ_F here is a *standard deviation*, propagated from measurement. It is not the side-flow ratio $\sigma(i)$, σ_0 of Table 2, which is a march variable. The rule: a subscripted σ with a measured quantity in the subscript (σ_F, and $\sigma_{F_{V1}}$, $\sigma_{F_{V3}}$, σ_{ratio} in the validation subsection) is always a dispersion; σ carrying a row index or the subscript 0 is always the march variable.

Exponent vs. lamina-width collision. β and γ are the cutoff-law *scaling exponents* (log–log slopes; symbolic $\frac{1}{2}$ and 1; empirical plateau fits ≈ 0.51 and ≈ 1.02). These must not be confused with the Inglis first-streamline *lamina width*, which this appendix writes as b (Inglis writes $D_c = 2\beta$; rendered $D_c = 2b$ here, Table 1). The prototype implementation overloads the name `beta` for both (`scaling.beta` for the exponent, `Dc=2*beta` for the width); this appendix disambiguates them as β (exponent) and b (width). The exponents are the falsifiable functional-form test of the cutoff law: a single global power law over the full range is rejected at reduced $\chi^2 \approx 30.6$, while the plateau exponents lie within the confidence intervals of $\beta = \frac{1}{2}$, $\gamma = 1$, surviving digitization noise.

A.4 Cutoff-Law Prefactor Decomposition (selectivity axis)

The selectivity axis is governed by the cutoff law

$$d_{\text{crit}} = \kappa_{\text{app}}\, w_c \sqrt{f_{\text{gap}}/3}, \qquad \delta = w_c \sqrt{f_{\text{gap}}/3}, \tag{1}$$

in which δ is the leading term of the exact dividing-streamline inversion of the skim cubic $f_{\text{gap}} = 3x^2 - 2x^3$ (Inglis ε-cubic, [13, p. 656]; $x = \delta/w_c$), and κ_{app} is the *apparent* prefactor that maps the idealized skim width onto the measured cutoff diameter. κ_{app} **is a plateau-region constant, not a universal one.** It is constant only for $\widetilde{d} \geq 0.035$; below that the prefactor becomes a function of the cutoff itself and Eqn 1 becomes implicit. Every use of the numerical value 4.9138 in this appendix is a plateau-region use. The prefactor factorizes as

$$\kappa_{\text{app}} = \kappa_{\text{steric}} \cdot K_{\text{in-plane}} \cdot K_{\text{depth}} \cdot \sqrt{c} \cdot C = 2 \cdot 1 \cdot R(\alpha) \cdot \sqrt{c} \cdot C. \tag{2}$$

Table 4: Cutoff-law prefactor factors. All strictly dimensionless. κ_{steric}, $K_{\text{in-plane}}$ and the depth factor K_{depth} are determined from first principles or single-gap CFD; the product $\sqrt{c}\,C$ is the single remaining calibrated (open) quantity.

Symbol	Name	Value / defining relation	Status	Source
κ_{app} ($\equiv \kappa_g$)	Apparent cutoff prefactor, cubic branch. ESTABLISHES	$d_{\text{crit}} = \kappa_{\text{app}} w_c \sqrt{f_{\text{gap}}/3}$; *supplied* $\kappa_{\text{app}} = 4.9138$, 95% CI $[4.75, 5.08]$, CV = 4.0%, plateau $\widetilde{d} \gtrsim 0.035$. *Refit* from the digitised points gives 4.913717814706706 (= 4.9137 to 4 d.p.); the two agree to four significant figures and differ by 1.7×10^{-5} relative. Use 4.9138 in prose and equations; use 4.9137 only where the fit itself is the subject	FITTED	[1, p. 4500, Fig. 3] digitization; [9]
$\kappa_{F_{\text{lam}}}$	Prefactor paired with the finite-aspect-ratio lamina kernel F_{lam}	$\kappa_{F_{\text{lam}}} = \kappa_g \sqrt{\overline{c_2(\alpha_{\text{cal}})}/3} = 5.798$ at the *assumed* 80 µm calibration depth. **The kernel F_{lam} is exact; the scalar $\kappa_{F_{\text{lam}}}$ is not.** It is an array-level transferred prefactor obtained from the cubic-branch value using the mean c_2 over an assumed calibration set, so it inherits an unreported depth. **Not interchangeable with** κ_{app}; see the pairing rule below and the α_{cal} entry	DERIVED from an EXTERNAL input	This work, computed from [9] (`c2`, `plateau_kappa`)
κ_{steric}	Steric factor	**= 2**: cutoff radius = lamina width ⇒ cutoff diameter = 2×lamina ($D_c = 2b$; b is defined in Table 1)	DERIVED, corroborated	[13, p. 656]; [5, p. 917] main text; [9] (`KAPPA_STERIC`)
$K_{\text{in-plane}}$	In-plane funnelling factor	= 1.000 (single-gap CFD Runs A/B/C; no in-plane aperture funnelling)	NUMERICAL	[7]; [9] (`K_INPLANE`)
$K^{\text{th}}_{\text{depth}}(\alpha)$	Depth factor, *analytic*	$K^{\text{th}}_{\text{depth}} = \sqrt{3/c_2(\alpha)} = \lim_{x\to 0} \delta_{3D}/\delta_{2D}$ from the rectangular-duct law. 0.8879/0.9500/0.9794 at $\alpha = 0.80/1.40/2.69$	DERIVED	[6, p. 875]; [9] (`c2`, `K_depth`)
$K^{\text{CFD}}_{\text{depth}}(\alpha)$	Depth factor, *CFD*	Single-gap CFD: 0.891/0.945/0.981 at the same three α. Agreement $\lvert K^{\text{CFD}}_{\text{depth}} - K^{\text{th}}_{\text{depth}}\rvert / K^{\text{CFD}}_{\text{depth}} \le 0.6\%$ (actual 0.349/0.530/0.160%, i.e. at most 0.531%, the largest rounded up to a bound, at $\alpha = 1.40$; the denominator is the CFD anchor, matching the gated `cfd_kdepth_obs`)	NUMERICAL	[7]; [9]
α_{cal}	Calibration aspect-ratio array	(0.8000, 0.6400, 0.5333) at $w_c = 100/125/150$ µm *assuming* $h_{\text{cal}} = 80$ µm. The depth is **unreported**: a genuine free parameter, and an EXTERNAL input rather than a computed quantity. Two distinct sensitivities follow and must be named apart; see below	EXTERNAL, undischargeable	This work (assumed); the assumption is recorded in [9]
c	Per-device over-extraction factor. COMPETES with the finite-particle account for the same residual	Enters as $\sqrt{c}$; the cumulative per-gap over-extraction tied to gap count N. *Lower-case c under the radical is not the finite-particle factor C below*: see the collision note	PROPOSED	[9]
C	Cumulative finite-particle factor	Body-averaged finite-sphere displacement vs. point streamline; *open*, to be computed by a finite-body method of the MOPSA class ([15, p. 2]). See the three-layer note below: the *mechanism* is established, the *method class* is available, the *factor* is this work's parameterization	PROPOSED, uncomputed	[17, p. 359] (mechanism); [15, p. 2] (method class)
$\sqrt{c}\,C$	Open residual (two factors, three gaps). ESTABLISHES the cutoff magnitude; its *split* does not	$K_{\text{depth}}\sqrt{c}\,C = \kappa_{\text{app}}/\kappa_{\text{steric}} = 2.4569$ (depth-folded, α-independent); $\sqrt{c}\,C = \kappa_{\text{app}}/(\kappa_{\text{steric}} K^{\text{CFD}}_{\text{depth}}) = 2.7574/2.5999/2.5044$ over $\alpha = 0.80/1.40/2.69$, depth-separated. *Never print the two conventions as one range*	FITTED / PROPOSED	[9]

c/C collision. Lower-case c is the per-device over-extraction factor and upper-case C the cumulative finite-particle factor. They are different quantities that differ only by case, and they appear together in a single product, $\sqrt{c}\,C$, which is the most misreadable arrangement possible. The radical is the only typographic

cue: c is *always* under it, C *never*. When either appears alone, restate which it is.

One open residual, two factors, three gaps. The residual is *derived from a fitted prefactor*, not measured: κ_{app} is calibrated on the Gifford plateau [1, 9], and published single-lineage data fix the product $\sqrt{c}\,C$ but not its split into c and C [9]. Within the model's premises the transport lemma makes $\kappa_{\text{steric}} = 2$ exact, so the residual $\sqrt{c}\,C$ must arise where a premise fails; the three model gaps the decomposition tracks are the flat-lamina idealisation, the wall porosity $\phi = G/\Lambda$, and finite-particle displacement [9]. Yet the factors number two: C is the finite-particle displacement alone (the anti-Zweifach–Fung attraction to the low-flow branch [17]), and because the decomposition $\kappa_{\text{app}} = \kappa_{\text{steric}} K_{\text{in-plane}} K_{\text{depth}}\,\sqrt{c}\,C$ closes by construction ($\kappa_{\text{steric}} = 2$, $K_{\text{in-plane}} = 1.000$), $\sqrt{c}\,C$ is the entire residual [9], so $\sqrt{c}$, the per-gap over-extraction, is that residual divided by C and carries the flat-lamina and porosity contributions together without isolating either. This grouping follows from the closure, not from any statement in the record, which names porosity as a source but assigns it to no factor [9]; neither the single-gap solve nor the yield identity resolves the split, which is what leaves the residual open [9].

Calibration depth (caveat). $\kappa_{\text{app}} = 4.9138$ is fit on the Gifford Fig. 3 array, whose caption reports only the three center-channel widths ($w_c = 100, 125, 150$ µm) and *no depth*. The value $\alpha = 0.8$ used for K_{depth} borrows the 80 µm depth of the *separate* Fig. 4 demonstration device at $w_c = 100$ µm; it is therefore an assumption, and K_{depth} in fact varies across the three calibration widths ($\alpha = 0.8000/0.6400/0.5333$ at $w_c = 100/125/150$ µm, giving $K_{\text{depth}}^{\text{th}} = 0.888/0.849/0.811$). The depth correction is therefore 11–19% across the calibration channels, not negligible. Crucially, K_{depth} is folded into the *apparent* prefactor as an interpretive factor and is *not* applied to the calibration data (which collapse more tightly on the parabolic δ), so it does not shift the fitted $\kappa_{\text{app}} = 4.9138$; it enters only the residual decomposition (depth-folded $\kappa_{\text{app}}/\kappa_{\text{steric}} = 2.4569$ vs. depth-separated $\sqrt{c}\,C$). Gifford states three device depths, none of them for this array: 80 ± 3 µm for the Fig. 4 device ([1, p. 4500, Fig. 4 caption]), 125 µm for the four Fig. 5 devices ([1, p. 4501, Fig. 5 caption]) and 150 ± 5 µm for the Fig. 6 platelet-enrichment device ([1, p. 4502, Fig. 6 caption]), so the borrowed 80 µm is the shallowest depth the paper reports. *The true array depth is not known and is not inferred here*; the consequences of that ignorance are quantified in the two α_{cal} sensitivities below.

The two α_{cal} sensitivities, named apart. Both are correct and they answer different questions. Do not call either "the β_{cal} swing" without saying which.

- $\Delta N_{\text{kernel|depth}} = +26.9\%$: the row-count inflation of an F_{lam} design over a cubic design at $h_{\text{cal}} = 80$ µm (against the cubic baseline $N = 2496$), falling to +3.9% at 150 µm. **Conditional on adopting F_{lam}**: if the kernel is never switched, h_{cal} does not enter N at all, because κ_g is fitted self-consistently on the cubic. This is the single dominant risk the design code flags: the one number that swings N by roughly 27%. *Provenance note: unlike every other figure in this subsection, these three numbers come from a calibration-depth sweep that is not reproduced in the derivation record and are not gated.*

- $\Delta_{\text{transfer}} = 20.3\%$: the within-array transfer bias, $c_2(0.53)/c_2(0.80) - 1$, for the array *as declared* $(0.80, 0.64, 0.53)$; 19.9% using the exact ratio 0.5333. The exact value is canonical in this appendix (a two-decimal input should not drive a four-decimal output); the declared-array figure is the one the verification registry gates and is labelled as such wherever it appears.

Steric criterion (adjudicated). The cutoff radius equals the extracted-lamina width (Dinh main text, $a < w$; [5, p. 917]), and the cutoff *diameter* is therefore twice the lamina width, matching Inglis's $D_c = 2\beta$ ([13, p. 656]; $D_c = 2b$ in this appendix's notation) and the Inglis–Herman restriction of particles "geometrically restricted away from walls by a distance equal to their radius" ([14, p. 1724]). Hence $\boldsymbol{\kappa_{\text{steric}} = 2}$. The Dinh Fig. 1 caption phrasing "about half the width of the extracted flow lamina" (equivalent to $\kappa_{\text{steric}} = 1$) is *deselected* in favor of the main text and Inglis. *This is an adjudication between two readings of the same paper, not a preference*: Dinh's main text and Fig. 1 caption are formally inconsistent, so one must be retracted, and the main text is selected because it agrees with the two independent statements of the same criterion in [13] and [14].

Finite-particle factor C: three layers, kept apart. The residual has physically motivated candidate explanations; its finite-particle component has not yet been computed directly. The evidence comes at three different strengths.

1. *Established physical mechanism.* A finite sphere near a bifurcation is drawn toward the *low-flow-rate* branch, the anti-Zweifach–Fung effect (Doyeux *et al.*, [17, p. 359]: the attraction is "directed towards

the low flow rate branch"), which in CIF is the side channel. Finite particles are therefore over-extracted relative to a point-streamline model, so the streamline that just excludes a particle is narrower than its radius and the measured cutoff diameter exceeds the idealized steric value 2δ. This is the physical content of $C > 1$.

2. *Available computational methodology.* Finite-body trajectory models of the MOPSA class ([15, p. 2]) can represent particles as finite rigid objects rather than point tracers; the paper "treats the particle as a two-dimensional rigid circular object instead of a single point."

3. *The factor C itself.* A dissertation-specific parameterization of the remaining finite-particle contribution, deferred to finite-particle work. It is *not* independently established by either [15] or [17]; those support the mechanism and the method, not this multiplicative form.

Its sign, mechanism and scale are corroborated on an unrelated device family in Section A.11.

Foot region and retraction. For $\widetilde{d} < 0.012$ the collapse degrades to a *sub-steric, recovery-suppressed* foot ($\kappa_{\text{app}} \to 1.34$, below the steric floor $\kappa_{\text{steric}} K_{\text{depth}}^{\text{CFD}} \approx 1.782$). *Both* 4.91 *and* 1.34 *are binned means, not extremes*: 4.91 is the mean of the $n = 8$ plateau bin ($\widetilde{d} \geq 0.035$) and 1.34 the mean of the $n = 3$ foot bin ($\widetilde{d} < 0.012$). The fifteen individual points span 1.20 to 5.20; that span is a different statistic and must not be quoted as the ramp. Depth cannot account for the foot because $K_{\text{depth}} \in [0.81, 0.98]$ cannot move κ_{app} from 4.91 to 1.34.

Scope of the evidence, stated explicitly. The foot bin is composed of three points at $d = 1.01$ µm (Gifford Fig. 3 rigid beads, $w_c = 100/125/150$ µm), roughly 1 µm particles, well below the 2.8–4.8 µm platelet cutoffs this framework targets, so the foot lies outside the regime the dissertation's claims cover. The corroborating observation comes from a *different* device and a *different* particle class: Dinh report that "recovery of the smallest 2 fL PLTs ($a \approx 1$ µm) in the filtrate was consistently lower than the percent of extracted fluid volume" ([5, p. 917]), and attribute it to "crowding of the flow laminae near the filtration gaps by retained" platelets, which "impeded the free outflow of the smallest PLTs" ([5, p. 917]). That is a concentration-dependent crowding effect in a platelet suspension, so its bearing on the dilute rigid-bead foot is by analogy of sign only. Recast in the language of this decomposition it corresponds to an effective sampled lamina narrower than nominal ($\delta_{\text{eff}} < \delta_{\text{nominal}}$), so that $\kappa_{\text{app}} = \kappa_{\text{steric}}(\delta_{\text{eff}}/\delta_{\text{nominal}})$ falls below steric; the recasting is this work's inference, the crowding mechanism is the source's, and the two are not rival accounts. Platelets are discoid and outside the rigid-sphere envelope (Section A.5), so the Dinh observation is offered as a mechanistic analogy of the same sign, not as evidence within scope, and no cutoff number is drawn from it. **The knee and foot mechanisms remain open**: the present data establish that the prefactor is size-dependent, but not which physical account of that dependence is correct.

The f_{eff} retraction is narrow. A *size*-dependent effective flow fraction $f_{\text{eff}}(d)$ remains a competing account of the knee, stated in its strongest form and not resolved here: the published data do not discriminate between it and the finite-particle account, because both predict the same sign and the same qualitative size dependence. It is *not* retracted. The cutoff law and the yield law are independent of which account is correct, since neither account enters them; only the *interpretation* of the residual turns on it. Retracted: $f_{\text{eff}}(N)$. Open and unresolved: $f_{\text{eff}}(d)$.

Scaling equivalence. Eqn 1 is equivalent to $D_c \propto w_c\sqrt{f_{\text{gap}}}$ with proportionality constant $\kappa_{\text{app}}/\sqrt{3} \approx 2.8370$; the 1/3 is carried explicitly inside the radical so that $\kappa_{\text{app}} = 4.9138$ is unambiguous.

Kernel–prefactor pairing rule. κ_{app} is meaningless without the kernel it was fitted with. Two kernels are in use: the leading-order skim branch g : $\delta = w_c\sqrt{f_{\text{gap}}/3}$ with prefactor $\kappa_g = 4.9138$; and the exact lamina-flux kernel F_{lam} with prefactor $\kappa_{F_{\text{lam}}}$. Declare a pair and use *that kernel at all three call sites*: the inversion, the schedule, and the scale $d_{\text{crit}} = \kappa_{\mathcal{K}}\delta$.

Conversion. Using the *array-mean* c_2 over the calibration widths:

$$\kappa_{F_{\text{lam}}} = \kappa_g\sqrt{\overline{c_2}/3} = \kappa_g/\sqrt{3/\overline{c_2}}, \qquad \overline{c_2} = \tfrac{1}{3}\textstyle\sum_i c_2(\alpha_{\text{cal},i}), \tag{3}$$

with $\alpha_{\text{cal}} = h_{\text{cal}}/w_c = 0.8000/0.6400/0.5333$ at the *assumed* $h_{\text{cal}} = 80$ µm, giving $\overline{c_2} = 4.177$ and $\kappa_{F_{\text{lam}}} = 5.798$.

Two sources of numerical discrepancy in that formula. Neither is pedantry; both move the third significant figure.

1. *Do not average the depth factors.* $\sqrt{3/\overline{c_2}} = 0.8475$ is an *effective* depth factor derived from the mean c_2. The arithmetic mean of the three K_{depth} values gives $\kappa_{F_{\text{lam}}} = 5.7862$ instead, 0.21% off, because the mean of the square roots is not the square root of the mean.

2. *Do not evaluate at one aspect ratio.* Using $K^{\text{th}}_{\text{depth}}(0.80)$ alone gives 5.5342, 4.6% low. The conversion is an array quantity.

What crossing costs, computed. At the Dinh geometry ($h = 140$ µm, $f_{\text{gap}} = 1.04 \times 10^{-4}$). Throughout this table δ_g and δ_F are the *exact* inversions of the cubic and of F_{lam} respectively, so that the two branches are compared like for like, and the percentages are computed on those exact widths at full precision, not from the rounded cells. The leading term $w_c\sqrt{f_{\text{gap}}/3}$ of δ_g is not used here: substituting it would give 2.170/1.504 for the first column, 0.2% below the tabulated 2.174/1.507, and would raise the kernel-choice figures above the tabulated +14.1%/ + 15.6%.

station	$\alpha = h/w_c$	$\kappa_g\delta_g$	$\kappa_{F_{\text{lam}}}\delta_F$	$\kappa_{F_{\text{lam}}}\delta_g$	$\kappa_g\delta_F$
inlet ($w_c = 75$)	1.867	2.174	2.482	2.565	2.103
outlet ($w_c = 52$)	2.692	1.507	1.743	1.779	1.477

The first two columns are the *self-consistent* pairs; the last two are crossings. Reading the inlet row: the kernel *choice* alone already moves the answer by +14.1% (2.174 vs 2.482 µm), rising to +15.6% at the outlet. Crossing a prefactor onto the other kernel's lamina shifts the result by a further +18.0% or −15.3%; one pair of figures covers both rows because the lamina width cancels, leaving the pure prefactor ratio $\kappa_{F_{\text{lam}}}/\kappa_g - 1$ and its inverse. The mismatch is not detected automatically and is therefore a potentially silent modelling error.

Which to use. The specification is directive, not neutral: until the calibration depth is measured, *use the cubic branch with* κ_g, which is self-consistent with the fit, and report the transfer-bias band as an uncertainty rather than applying a correction whose calibration cannot be verified. This dissertation's cutoff law (Eqn 1) and its entire validation path are on the cubic branch. The design code carries both pairs: its `design` function defaults to the cubic pair, and the physics layer that the design-map test marches (App. B, P-C8.2) runs on F_{lam}. The pairing rule is what keeps the two apart. See the α_{cal} entry for the row-count consequence of switching ($\Delta N_{\text{kernel|depth}} = +26.9\%$).

Rounding convention for α_{cal}. The array is declared in the design code as (0.80, 0.64, 0.53), but 0.53 is a two-decimal rounding of $h_{\text{cal}}/w_c = 0.5333$. This appendix uses the *exact* ratio: $\overline{c_2} = 4.177$ and $\kappa_{F_{\text{lam}}} = 5.798$. Under the declared rounding they are 4.182 and 5.802. The same split appears in the transfer bias: 19.9% exact against 20.3% for the declared array, which is the figure the verification registry gates. Both are correct for what they are; the exact ratio is canonical here and the declared-array figure is labelled as such wherever it appears.

A.5 Particle and Separation Variables

Table 5 gives the particle variables. The distinction that matters is between a volume-equivalent sphere and a real discoid platelet: the cutoff law returns the former.

Table 5: Particle-property variables, together with the one separation-performance quantity they set: the cutoff diameter. The filtrate fraction F and its variants are in Table 3, and the validation dispersions in the V&V table.

Symbol	Quantity	SI unit	Practical unit	Source
d_{crit}	*Critical (cutoff) diameter* (canonical; output of Eqn 1)	m	µm	[9]; [13, p. 656]
D_c, c.d.	Cutoff diameter (Xia/lineage symbol, $\equiv d_{\text{crit}}$); Xia introduces the abbreviation as "particles above a certain size (or 'critical diameter', c.d.)"	m	µm	[2, p. 2 (Sci Rep PDF pagination)]

Table 5 (continued)

Symbol	Quantity	SI unit	Practical unit	Source
a	Particle radius, as used in the *adopted* capture criterion $a < w$ (main text). *Not* the "effective radius" of the deselected Fig. 1 caption on p. 916; see the steric adjudication in Section A.4	m	µm	[5, p. 917]
$D = 2a$	Particle diameter (sphere)	m	µm	Standard
V_{sph}	*Sphere* volume, $V_{\text{sph}} = \frac{4}{3}\pi a^3$. Used for the equivalent-sphere conversion only; real platelets are discoid and this is *not* their volume formula	m^3	fL ($\equiv$ μm^3)	Standard
V_{12}	Dinh platelet-volume *classification threshold*, 12 fL: "cell volumes larger than 12 fL are commonly considered 'large"', the P-LCR cut used to define "> 12 fL recovery". *Not* a minimum retained volume	m^3	fL	[5, p. 913]

D/d collision: diameter-symbol reconciliation. d_{crit} is canonical for the cutoff law; D_c is its lineage synonym (Xia). Neither is to be confused with the deprecated depth symbol d (use h), nor with the hydraulic diameter D_h (Table 1). For a sphere $D = 2a$.

Two further case collisions, declared. V is a volume; lower-case v is a local fluid velocity (Table 2). V_{sph} and V_{12} are one quantity, a volume, read two ways. A particle radius is written a throughout and *never* r. Lower-case r is not used as a variable anywhere in this appendix, and reserving it keeps it clear of the resistances R_c, R_s, R_{gap} and their aliases (Table 2).

Volume convention. 1 fL = 1 μm^3 (by definition: 1 femtoliter = 10^{-15} liter = 10^{-18} m^3 = $(10^{-6})^3$ m^3). For comparison with the rigid-sphere model, 12 fL corresponds to an equivalent-sphere diameter $D = 2\,(3V_{\text{sph}}/(4\pi))^{1/3} \approx 2.84$ µm. *This is a geometric conversion, not a measured platelet diameter.* Dinh report the actual *discoid* platelet as $2a \approx 4.8$ µm at 12 fL, which in their own terms is the major/equatorial axis of an oblate spheroid, i.e. the in-plane one ([5, p. 917]); the $\sim 1.7\times$ gap between the two reflects the sphere idealization and therefore marks a shape-related *limitation* of applying the rigid-sphere model to platelets. The model is in scope; real platelets are not validated against it. A cutoff diameter predicted by Eqn 1 is therefore an *equivalent-sphere* diameter; for discoidal platelets it under-states the physical in-plane diameter by this factor.

A.6 Validation and Uncertainty Quantities (V&V 20, yield axis)

Table 6 lists the validation and uncertainty quantities with the COMSOL and reproducibility diagnostics. The framework is ASME V&V 20 [20]: the comparison error $E = S - D$ is judged against a combined validation uncertainty $u_{\text{val}} = \sqrt{u_{\text{num}}^2 + u_{\text{input}}^2 + u_D^2}$, of which the experimental term $u_D = 0.033$–0.036 is by far the largest. Each Dinh variant was solved at two mesh levels a ~5-fold refinement apart (V1: 115,904 and 24,783 elements; V3: 104,206 and 21,448), fully converged in four Newton iterations, *these are the convergence-study meshes*; the production V3 solve used a slightly different mesh of 104,651 elements, so the two V3 figures are different meshes rather than a discrepancy, and the reported F is identical at both [8, COMSOL reports; Block 19]. *That bounds* u_{num} *at* 5×10^{-5} *rather than measuring it*: five significant figures is the precision to which the flow tables are printed, so the true discretisation error is smaller and unresolved. Even as a bound it is at least 660× below u_D (660× at V1, 720× at V3).

The remaining term, u_{input}, is bounded by fabrication tolerance in two independent components; the boundary conditions do not contribute separately, because F is a ratio of filtrate flow to inlet flow, in which the imposed inlet rate and the outlet pressure datum both cancel identically. *Out of plane*, the channel-depth tolerance shifts F by at most $0.031\,u_D$ at V1 and $0.015\,u_D$ at V3 over ±5 µm [8, §8.6]. *In plane*, F is likewise a channel-resistance *ratio* in which isotropic scaling of the widths and lengths cancels to leading order, the only residual being the aspect-ratio bracket of the resistance law (Eqn 6), which is second order in the depth-fixed width perturbation and stays well below the out-of-plane term, so the 0.05% bound below already accommodates it. So u_{val} exceeds u_D by at most 0.05% (0.048% at V1, 0.011% at V3): to the two

decimals given E/u_{val} equals E/u_D, so the reported distances are governed by the choice of u_D scale, not by the small u_{num} or u_{input} terms. See the SD-vs-SEM note for that choice.

Table 6: Quantities used in validation, COMSOL simulation, and reproducibility infrastructure, including the ASME V&V 20 uncertainty terms.

Symbol	Quantity	Unit	Source
$\lvert\Delta F_{\text{1D}}\rvert$	Absolute difference: closed-form F vs. recursion endpoint. *A one-dimensional self-consistency check, not the mesh-to-mesh difference*	—	[8, §8.4]
u_{val}	Combined validation uncertainty, $u_{\text{val}} = \sqrt{u_{\text{num}}^2 + u_{\text{input}}^2 + u_D^2}$; evaluates to $\approx u_D$ here	—	[20]; [7]
u_{num}	Numerical (discretisation) uncertainty. *Bounded, not measured*: $\leq 5 \times 10^{-5}$, the reporting precision at which F is unchanged across two mesh levels a ∼5-fold refinement apart, i.e. at least 660× below u_D	—	[7]; [8, COMSOL reports]
u_{input}	Input uncertainty (geometry, boundary conditions). $\leq 0.031\,u_D$ from the channel-depth tolerance over ±5 µm; isotropic in-plane fabrication error cancels to leading order in the resistance ratio	—	[8, §8.6]
u_D	Experimental (data) uncertainty, reported here as the published SD (a conservative choice; see the SD-vs-SEM note). V&V 20's u_D for a mean of n has random component SEM = SD/$\sqrt{n}$, reported alongside	—	[5, p. 919]
E, S, D	Validation comparison error $E = S - D$, simulation value S minus data value D. *All three letters are reused*: see the collision note below	—	[20]
$T_{\text{PE},1/2}$	PE half-life: gap rows for outlet pressure mismatch to halve	gaps (count)	[8, §8.7]
α_{3D}	3-D outlet resistance-ratio correction $(R_c/R_s)_{\text{app}}/(R_c/R_s)_{\text{Yang}}$ (yield-side). *Not* the channel aspect ratio α of Table 3	—	[7]; [8, §8.8]
$\sigma_{F_{V1}}, \sigma_{F_{V3}}$	One-SD experimental F uncertainty (Dinh, n=4, panel B). A subscripted σ naming a measured quantity is a dispersion, never the side-flow ratio $\sigma(i)$ of Table 2	—	[5, p. 919]
σ_{ratio}	Propagated uncertainty on multi-variant ratio	—	[5, p. 919]; [8, §15.4]
$N_{\text{elem}}, N_{\text{DOF}}$	COMSOL mesh element / degrees-of-freedom count	count	[7]; [8, Block 19]
ϵ_{conv}	Newton convergence residual tolerance. *Not* the DLD row-shift fraction ε of Table 3; the two epsilons are one glyph	—	[7]
$N_{\text{validators}}, N_{\text{stamps}}$	Validator-suite assertion / regression-stamp count	count	[9]

$E = S - D$ collision: three reused letters in one formula. The V&V 20 comparison error is written $E = S - D$, and each letter already denotes something else in this appendix. S here is the *simulation* value of F; $S(i)$ in Table 2 is the normalised side-channel flow along the march. D here is the *data* value of F; $D = 2a$ in Table 5 is a particle diameter, and D_c, D_h are the cutoff and hydraulic diameters. The rule: inside $E = S - D$, and nowhere else, S and D are the V&V 20 simulation and data values; they are dimensionless and always appear in that formula, or as the reported ratio E/SD, in which the roman SD is the standard-deviation abbreviation and not the italic pair S, D. Elsewhere they keep their table meanings.

α_{3D} collision. $\alpha = h/w_{\text{ch}}$ is the channel aspect ratio and α_{cal} its three-element calibration array (Table 3); both are geometric. α_{3D} is neither: it is a *resistance-ratio* correction on the yield axis, dimensionless but unrelated. The subscript is the only cue, so restate it wherever the two could be read together.

ε/ϵ collision. LaTeX offers two epsilons and they are one glyph to a reader. ε is source notation only, the DLD row-shift fraction quoted from the deterministic-lateral-displacement literature (Table 3); it is never a CIF quantity. ϵ_{conv} is this work's Newton convergence tolerance. The rule: a bare epsilon is always the source's, and a subscripted one is always this work's.

***u*-symbol collision.** The letter u carries three unrelated meanings, two of them inherited and not negotiable. $u(y)$ and $u_{\max}$ are fluid velocities (Table 2, standard fluid mechanics). u_{val}, u_{num}, u_{input} and u_D are ASME V&V 20 uncertainties (this table, the standard's own notation). The third, $\widetilde{d} = d_{\text{crit}}/w_c$, was moved off the letter entirely and is rendered $\widetilde{d}$ (Table 3) precisely because u_d and u_D differ by subscript case alone.

N-symbol collision. Bare N is the total gap-row count (Table 1) and is load-bearing in the schedule and cutoff laws. Every other N in this appendix carries a subscript naming what it counts: N_{elem}, N_{DOF}, $N_{\mathrm{validators}}$, N_{stamps}, all of which are infrastructure diagnostics. A bare N is never a count of anything but gap rows.

σ collision. A subscripted σ that names a *measured* quantity is a dispersion, never a flow ratio: $\sigma_{F_{V1}}$, $\sigma_{F_{V3}}$ and σ_{ratio} are experimental standard deviations (this table), while $\sigma(i)$ and its inlet value σ_0 are the normalised side-channel flow along the march (Table 2). The two are told apart by subscript *content*, not case: a bare or row-indexed σ is always the flow ratio, a device- or ratio-subscripted σ is always a standard deviation.

SD vs. SEM. Dinh Fig. 4 reports *mean* ± *standard deviation* (n=3 panel A; n=4 panel B; [5, p. 919]). The F-validation is the two-device comparison in panel B, so the datum D is a mean of n=4 (the V1 recovery $50.5 \pm 3.3\%$ is also reported in panel A at n=3; the panel B quadruplicate is the one used here). ASME V&V 20 defines u_D as the standard uncertainty appropriate for D and states that it "includes all effects of averaging, includes all random and systematic uncertainty components" ([20, §1-6]); for a mean of n the random component is therefore the standard deviation of the mean, SEM = SD/$\sqrt{n}$, so the distance measured in units of the standard's own random uncertainty is E/SEM.

The published SD **is nevertheless retained here as** u_D, as a deliberately *conservative* choice on two grounds: it over-covers that random component by $\sqrt{n}$, and it stands in for the systematic experimental uncertainty that V&V 20's u_D formally includes but that Dinh do not separately report. Both distances are given below so the reader can see the choice rather than inherit it. Under the stricter SEM scale the closed form and the 3-D solve remain within 2σ (the largest, V3 COMSOL, is -1.16σ); on the raw-SD scale they are within 1σ.

Worked validation numbers (Dinh V1, V3).

	$F_{V1}^{\mathrm{closed}} = 0.5592,$	$F_{V3}^{\mathrm{closed}} = 0.7189$	(closed form, no fit)
	$F_{V1}^{\mathrm{COMSOL}} = 0.5070,$	$F_{V3}^{\mathrm{COMSOL}} = 0.6281$	
	$F_{V1}^{\mathrm{exp}} = 0.505 \pm 0.033,$	$F_{V3}^{\mathrm{exp}} = 0.649 \pm 0.036$	(SD, n=4)
E/SD (1-D closed form) :	$V1 = +1.64,$	$V3 = +1.94$	
E/SD (3-D COMSOL) :	$V1 = +0.06,$	$V3 = -0.58$	
E/SEM (3-D COMSOL) :	$V1 = +0.12,$	$V3 = -1.16$	(within u_{val}).

What "calibration-free" claims. The calibration-free 1-D closed form is $+1.64\sigma$ (V1) and $+1.94\sigma$ (V3) above the measurement, consistent within 2σ but not a match. The 3-D outlet solve is closer: within 1σ on the SD scale and within 2σ on the stricter SEM scale ($+0.12\sigma$ / -1.16σ), and $\alpha_{3\mathrm{D}} = 0.811/0.660$ quantifies the outlet-resistance correction required to reconcile the two. The claim is that the yield law requires *no device-specific fitting*, not that the one-dimensional form reproduces the measurement. State both distances wherever one is stated. *Whether this yield-side residual and the cutoff-side residual of Section A.4 share a common three-dimensional origin is an open question and is deferred to future work*; nothing in this appendix turns on the answer, and the two are treated throughout as corrections on independent axes.

A.7 Selectivity-Axis Validation (predicted vs. measured cutoff)

The yield axis is validated directly against a measured mean with a stated dispersion (Section A.6). The selectivity axis is independently checked using published devices that took no part in the κ_{app} fit, **although the available checks are materially weaker than the yield validation**: the Dinh check is genuinely held out but resolves only a wide interval bracket, and the Xia check is not independent of the calibration at all. **Neither delivers a sharp held-out validation**, and both are reported with the reasons why.

Dinh 2024: weak interval-consistency check. The framework predicts a cutoff band [1.25, 2.17] µm for the Dinh device: the registered band, recomputed by the derivation record [9] and carried throughout the formal argument. **Its width is an ambiguity, not a gradient**, and the distinction matters because it is easy to read the band as a cutoff that varies down the device. It does not, or at least the lineage design says it should not.

Why the taper does not move the cutoff. The centre channel narrows for a reason unrelated to selectivity: Xia introduced the taper "in order to reduce the maximum amount of shear stress experienced by particles

within the device" ([2, p. 4 (Sci Rep PDF pagination)]), which is what the RS in RS-CIF stands for. Narrowing w_c alone would shrink the lamina and drag the cutoff down with it, so the lineage ramps the extraction to compensate: "In order to maintain a consistent c.d. along the length of an RS-CIF device, the value of f_{gap} can no longer be kept constant and must be scaled relative to" the narrowing ([2, p. 9 (Sci Rep PDF pagination)]). That scaling is Xia's eqn. (1), stated here in its *corrected* form, $f_{\text{gap}}(i) = (w_c(\text{ref})/w_c(i))^2\, f_{\text{gap}}(\text{ref})$. The 2016 printing ([2, p. 10 (Sci Rep PDF pagination)]) inverted the two widths, and the 2022 Correction records that the article "contains an error in equation 1, where the terms" $w_c(i)$ and $w_c(\text{ref})$ "are inverted" ([3, p. 1]). **The direction is load-bearing here**, so the version matters: with the widths inverted, $\delta \propto w_c(i)^2$ and the lamina shrinks *faster* than the bare taper, which is the opposite of compensation. In the corrected form it cancels the taper exactly:

$$\delta(i) = w_c(i)\sqrt{\tfrac{f_{\text{gap}}(i)}{3}} = w_c(i)\,\frac{w_c(\text{ref})}{w_c(i)}\sqrt{\tfrac{f_{\text{gap}}(\text{ref})}{3}} = w_c(\text{ref})\sqrt{\tfrac{f_{\text{gap}}(\text{ref})}{3}},$$

independent of w_c. A tapered device is designed to hold one cutoff, not a range of them.

What the band's width therefore is. For the bracket device Dinh states a single initial $f_{\text{gap}} = 1.04\times10^{-4}$, one of three initial values its designs span ([5, p. 917]), and describes such a value as one of "the initial values for the width of the middle channel and the filtration fraction" ([5, p. 915]); the paper states that the later values are "iteratively adjusted with each subsequent gap i … to maintain the desired size cut-off throughout the device" ([5, p. 916]), citing Xia 2016 for the method, but does not print them. The band is computed holding it constant, which makes $\delta \propto w_c$ and spreads the prediction across the taper. Of the band's 1.74× width, 75/52 = 1.44× is that spread and only 1.21× is the near-knee κ_{app} ambiguity, where the fitted prefactor has ramped off the plateau κ_g to a local 4.08. Under the Xia-scaled reading, with f_{gap} pinned at the initial station, the lamina, and with it the cutoff at the plateau prefactor, is a single value and the band collapses to [1.80, 2.17], the κ_{app} ambiguity alone. *The wider band is retained because it is the conservative envelope*: it contains the Xia-scaled reading, which is the one Dinh describes, in its corrected form (Appendix B, C5), and taking the narrower one would assume that a schedule the paper describes but does not print meets its stated intent exactly.

The measured rigid-bead bracket is (1.0, 4.7) µm (Dinh ESI Fig. S1: 1 µm beads track the fluid, 4.7 µm beads are retained). The predicted band lies *inside* the measured bracket on either reading. **This is a weak test and must be reported as one:** the bracket is 4.7× wide, wider than the prediction it is supposed to discriminate; the device operates near the knee, where held at its initial gap fraction the self-consistent cutoff sits at the sub-steric foot, beneath the bracket, so the band is anchor-contingent; and the band is doubly sensitive to the assumed calibration depth and to the unresolved f_{eff}. It is a consistency check, not a sharp validation.

Xia 2016: cross-paper consistency check. *Recalculation* from three Xia devices, using their stated cutoff and f_{gap} values, gives $\kappa_{\text{app}} = 5.03/5.16/5.32$ (mean 5.17) at 140 µm depth against the Gifford plateau 4.9138 fitted at an assumed 80 µm, which agrees to 5.2% across papers and depths. All three device triples are quoted data: OR-7 at $f_{\text{gap}} = 5.8\times10^{-4}$ for a 7 µm cutoff, and RS-8 and RS-9 at 7.2×10^{-4} and 8.6×10^{-4} for 8 and 9 µm, all at $w_c(\text{ref}) = 100$ µm, are each stated verbatim in the source.

This anchor is not independent, and the agreement is not diagnostic. Two separate reasons, and the second is the stronger one. First, Xia reads its cutoff from the same Gifford figure the plateau was fitted from, and Gifford is a Xia co-author, so the anchor and the fit share a figure rather than merely a literature. Second, and decisively: the 5.2% by which the recomputed mean exceeds 4.9138 is numerically the depth correction itself. Xia's devices are 140 µm deep against the plateau's assumed 80 µm, and $1/K^{\text{th}}_{\text{depth}}(1.40)-1 =$ 5.3%, which is the same correction as the 5.2% agreement above (the two printed figures differ only because they fall either side of a rounding boundary at full precision), so 4.9138/0.9500 reproduces the recomputed mean to three significant figures. The observed agreement is therefore *equally consistent* with a depth-robust prefactor and with a depth correction of exactly that size: it discriminates between them not at all. The recalculation is real and rests on quoted data, but it is reported here as a *lead*, not as a cross-paper agreement, and the argument records `xia_anchor_unverified` as open.

Numerical uncertainty here is bounded differently. The single-gap CFD behind $K_{\text{in-plane}}$ and the $K^{\text{CFD}}_{\text{depth}}$ anchors has no h-refinement study. Its resolution rests instead on agreement with the exact analytic duct law to $\leq 0.6\%$ at every point and aspect ratio (the depth factors themselves agree to 0.349/0.530/0.160%), supported by mass conservation to 0.015% and an export-grid check that moves δ by

$\leq 0.1\%$ under an $8\times$ coarsening of the *sampling* pitch. A sub-micron skim that lands on an analytic value to six parts in a thousand cannot be under-resolved, so the discretisation error on δ is bounded near 0.6%: some $5\times$ below the $\pm 3.4\%$ confidence interval on κ_{app} it feeds, and finer than the $4.7\times$-wide measured bracket above can resolve.

Fitted-prefactor uncertainty. κ_{app} carries a 95% CI $[4.75, 5.08]$ and CV = 4.0% over the $n = 8$ plateau devices. The single global power law is rejected at reduced $\chi^2 \approx 30.6$, robust across three independent re-digitizations; that rejection is the falsifiable content of the functional-form test. *The plateau-only fit is itself* $\chi^2_{\mathrm{red}} \approx 3.8$*, which is not a good fit in absolute terms*; the claim is relative, that 30.6 against 3.8 is a factor-8.0 degradation which rejects the global law.

A.8 Validity-Envelope and External Inputs

The design framework operates inside a declared envelope. These quantities are the boundary conditions on that envelope, and several are EXTERNAL: supplied, and undischargeable by any validator in this work. An appendix that catalogued the framework without them would omit the conditions under which its own claims hold.

Table 7: Validity-envelope quantities and external inputs.

Symbol	Role	Status	Source
u_{lo}	Plateau guard: every row must satisfy $\widetilde{d} = d_{\mathrm{crit}}/w_c \geq u_{\mathrm{lo}}$, keeping the design above the knee. *A threshold on $\widetilde{d}$, not a velocity and not a V&V uncertainty*; see the u note below	EXTERNAL	This work (supplied)
u_{ff}	Free-flow guard on the reduced size ratio $\widetilde{d}$, same letter caveat	EXTERNAL	This work (supplied)
η	Realized-yield factor $\eta = 0.903$: a three-dimensional correction on the *flow split*. **Not** an external free parameter: its shortfall $1 - \eta = 9.7\%$ is the same $\sim$10% one-dimensional-versus-three-dimensional residual on F that the outlet solve closes, so it is a computed magnitude recorded here rather than a supplied constant. *A single realized-yield magnitude, not a per-variant fit*: the actual residual is 9.3% at V1 and 12.6% at V3, so 9.7% is representative and matches V1 closely rather than holding exactly at both. **Not** the lamina model gap either, which lives in κ_{app}; the two act on different axes and must not be traded against one another	MEASURED (corroborated by the 3-D COMSOL outlet solve to within 1σ, not derived by the one-dimensional treatment)	[9], `uncheckable_eta`
ϕ_{cal}	Calibration wall-porosity input, distinct from $\phi = G/\Lambda$, which is *computed* from geometry; ϕ_{cal} is supplied and no validator discharges it. See the ϕ note below	EXTERNAL	This work (supplied); [9]
σ_0	Fictitious inlet side-flow in the mass identity $P(i) + 2S(i) = 1 + 2\sigma_0$. *Defined in Table 2*; repeated here only to record that it is supplied	EXTERNAL	Table 2
f_{soft}, γ_{GS}	Schedule-softening and gap-scale process parameters. f_{soft} is a *process* parameter and is not the filtrate fraction f_{gap}; see the f note below	EXTERNAL	This work (supplied)
$\mathrm{AR}_{\mathrm{max}}$, R_{min}	Fabrication limits: maximum feature aspect ratio and minimum inner-corner radius	EXTERNAL	This work (supplied)
c_{ent}, L_{eq}	Entrance coefficient and bend-equivalent length. *The link length ℓ_g is defined in Table 1* and is not redefined here	EXTERNAL	This work (supplied)
Three conditions a design must satisfy. Each is stated as a boundary rather than asserted to hold, because a real device can violate any of them; asserting they always hold would claim the device is always above the knee, which is false.			
Above-knee operation	If the device runs below the knee, κ_{app} ceases to be a constant and becomes a function of the cutoff itself, making the design relation implicit	BOUNDS	This work
Run-out match	A $3\times$ run-out mismatch collapses F from 0.519 to 0.265; the outlet lengths must be matched	BOUNDS	[9], `runout_mismatch`
Free row count	Pinning N over-constrains the problem: two targets meet one free variable, and the orthogonality of cutoff and yield is lost	BOUNDS	This work

u-symbol collision, extended. Table 6 records three meanings for the letter u: the fluid velocities $u(y)$ and $u_{\max}$, the V&V 20 uncertainties u_{val}, u_{num}, u_{input}, u_D, and the reduced size ratio, which was moved *off* the letter to $\widetilde{d} = \widetilde{d}$ precisely because u_d and u_D differ by subscript case alone. The guards u_{lo} and u_{ff} are a fourth use, and an awkward one: they are thresholds *on* $\widetilde{d}$, so they are named after the very symbol this appendix stopped using. They keep their names because they are the identifiers the design code and the specification use, and renaming them here would break that correspondence, but the reader should read u_{lo} and u_{ff} as bounds on $\widetilde{d}$ and never as velocities or uncertainties.

f-symbol collision. f_{gap} is the per-gap filtrate fraction and is the central dimensionless quantity of the yield axis (Table 3). f_{soft} is a schedule-softening *process* parameter with no flow meaning, and F with its variants is the cumulative filtrate fraction. Three f-shaped objects, distinguished by subscript and by case.

ϕ-symbol collision. ϕ_{cal}, the wall porosity *supplied* to the calibration, is distinct from $\phi = G/\Lambda$, which is *computed* from the geometry (Table 1), and and from the upper-case fold factor $\Phi(i) = r(w_s(i))/r(\text{GS})$ of the folded-channel geometry (Table 1): one letter, separated by subscript and by case.

c-symbol collision. The letter c carries three unrelated meanings. In the cutoff decomposition $\sqrt{c}\,C$ (Table 4) lower-case c is the per-device over-extraction factor and upper-case C the cumulative finite-particle factor, distinguished by case alone. c_2 is the near-wall coefficient of the velocity expansion (Table 3). And c_{ent}, in this table, is a dimensionless entrance-length coefficient with no relation to either. All three are dimensionless, so unit disagreement will not separate them; the subscript is the only cue.

L-symbol note. L_{eq}, the bend-equivalent length, is a member of the length family alongside L_{dev}, $L_c(i)$ and $L_s(i)$ (Table 1): all are lengths in metres and all carry a subscript naming what they measure.

On the EXTERNAL **label.** In this table it has the second of the two senses Section A.1 gives it: it marks a quantity this work supplies rather than derives, and which no validator in the record discharges. That is a statement about *this* record, not a claim that the quantity is underivable: ϕ_{cal} and the fabrication limits are physically measurable, and a later campaign that measured them would move them out of this table. The label is a boundary on what the present argument establishes.

A.9 Equations: Step-by-Step Unit Derivations

This section provides a dimensional *consistency check* for each load-bearing equation. The notation $[X]$ denotes "the SI unit of X." *Dimensional consistency is necessary, not sufficient*: it shows an equation is well formed, not that its physical assumptions hold. The assumptions are stated separately with each equation.

A.9.1 Gifford 2014 Eqn 1: flow recursion

$$Q_s(i) = Q_s(i-1) + f_{\text{gap}}(i) \cdot Q_c(i-1) \tag{4}$$

Dimensional analysis:

$$[\text{LHS}] = \text{m}^3\,\text{s}^{-1}$$
$$[\text{RHS}] = \text{m}^3\,\text{s}^{-1} + \underbrace{[f_{\text{gap}}]}_{1} \cdot \underbrace{[Q_c]}_{\text{m}^3\,\text{s}^{-1}} = \text{m}^3\,\text{s}^{-1}$$

[LHS] = [RHS]. ✓ *Index note:* Gifford writes Eqn 1 with no row index on f_{gap} (his constant-f_{gap} regime); the (i) here is the tapered-f_{gap} generalization. $Q_c(i)$ and $Q_s(i)$ are the flows following gap row i, as Gifford defines them, so $f_{\text{gap}}(i)$ is the fraction taken at row i from the central flow reaching it, $Q_c(i-1)$, which is Dinh's $Q_m(i)$ ([5]). Every equation in this appendix uses this labelling.

A.9.2 Gifford 2014 Eqn 3: continuity

$$\frac{\Delta P(i)}{\Delta P(i-1)} = 1 - 2f_{\text{gap}}(i) \tag{5}$$

Dimensional analysis: [LHS] = Pa/Pa = 1; [RHS] = $1 - 2 \cdot 1 = 1$. ✓The factor of 2 is bilateral extraction (each of two side channels takes f_{gap} of Q_c). *Index note:* Gifford's verbatim form carries no index on f_{gap}; the (i) is the tapered generalization.

A.9.3 Gifford 2014 Eqn 5: Yang series for hydraulic resistance

$$R = \frac{12\mu L}{w_{\mathrm{ch}} h^3}\left[1 - \frac{192\,h}{\pi^5 w_{\mathrm{ch}}}\sum_{n=1,3,5,\ldots}^{\infty}\frac{1}{n^5}\tanh\!\Big(\frac{n\pi w_{\mathrm{ch}}}{2h}\Big)\right]^{-1} \tag{6}$$

Dimensional analysis (convention $h \le w_{\mathrm{ch}}$): prefactor $[12\mu L]/[w_{\mathrm{ch}}h^3] = \mathrm{Pa\,s\,m/m^4} = \mathrm{Pa\,s\,m^{-3}}$; the bracket is dimensionless, so $[R] = \mathrm{Pa\,s\,m^{-3}}$. ✓

Attribution. Gifford's Eqn 5 carries citation [20] = Yang/Ündar/Zahn ([6]), whose Eqn 3 is the rectangular-duct series; Gifford's ref [19] (Sethu/Sin/Toner) is the *simplified* $R(w)$ treatment Eqn 5 improves on, not its source.

Limit checks. Parallel plates ($h \ll w_{\mathrm{ch}}$): series $\to 0$, $R \to 12\mu L/(w_{\mathrm{ch}}h^3)$. ✓Square channel ($h = w_{\mathrm{ch}}$): $R \to 28.4\,\mu L/h^4$ (Bruus 2008, [16]). Independently corroborated by the registry's square-duct anchor $12/B(1) = 28.4542$, computed from the duct series with no reference to [16]. ✓*Note:* the channel-width argument is $w_{\mathrm{ch}} \in \{w_c, w_s\}$, not the lamina width w.

A.9.4 Dinh 2024 p. 915: definition of f_{gap}

$$Q_{\mathrm{gap}}(i) = f_{\mathrm{gap}}(i)\cdot Q_c(i-1) \tag{7}$$

Dimensional analysis: $f_{\mathrm{gap}}(i) = Q_{\mathrm{gap}}(i)/Q_c(i-1)$, units $\mathrm{m^3\,s^{-1}/m^3\,s^{-1}} = 1$. ✓*Index note:* Dinh writes the flow reaching gap i as $Q_m(i)$ ([5]); in Gifford's labelling, used throughout this appendix, it is $Q_c(i-1)$.

Numerical sanity (Dinh V1): $Q_c \approx 30$ µL min^{-1}, $f_{\mathrm{gap}} = 1.04\times10^{-4} \Rightarrow Q_{\mathrm{gap}} = 3.12$ nL min^{-1} per gap per side.

A.9.5 Xia 2022 corrected scaling for matched cutoffs

$$f_{\mathrm{gap}}(i) = \left(\frac{w_c(\mathrm{ref})}{w_c(i)}\right)^2 f_{\mathrm{gap}}(\mathrm{ref}) \tag{8}$$

Dimensional analysis: ratio squared is dimensionless; both sides dimensionless. ✓At fixed cutoff: $f_{\mathrm{gap}}\,w_c^2 = \mathrm{const}$, i.e. $d_{\mathrm{crit}} \propto w_c\sqrt{f_{\mathrm{gap}}}$, consistent with the cutoff law of Section A.4. *This is the* ***corrected*** *form*: the 2016 printing inverted $w_c(i)$ and $w_c(\mathrm{ref})$, and the 2022 Correction records that the article "contains an error in equation 1, where the terms" $w_c(i)$ and $w_c(\mathrm{ref})$ "are inverted" ([3, p. 1]). The direction is load-bearing: inverted, the lamina would shrink *faster* than the taper instead of holding constant.

A.9.6 Skim cubic and dividing-streamline width (selectivity)

$$f_{\mathrm{gap}} = 3x^2 - 2x^3, \qquad x = \delta/w_c; \qquad \delta = w_c\left[\sqrt{f_{\mathrm{gap}}/3} + \tfrac{1}{9}f_{\mathrm{gap}} + \cdots\right] \tag{9}$$

Dimensional analysis: f_{gap} and x are dimensionless; the cubic balances $1 = 1$. ✓$[\delta] = [w_c]\cdot 1 = \mathrm{m}$. ✓This is the Inglis ε-cubic ([13]): the parabolic-profile flux fraction equals $3x^2 - 2x^3$. Inglis states the premise, "by assuming a conventional parabolic flow profile through the gap" ([13, p. 656]), and writes the result in monic cube-root form, $y^3 - \frac{3}{2}y^2 + \frac{\varepsilon}{2} = 0$ with $y = b/g$; multiplying by -2 and identifying $\varepsilon \leftrightarrow f_{\mathrm{gap}}$, $y \leftrightarrow x$ recovers $f_{\mathrm{gap}} = 3x^2 - 2x^3$ exactly. *The polynomial is a derived identity, not a quoted formula*, which is why the source-quote index anchors the premise rather than the cubic.

Accuracy of the leading inversion, with the band named. The leading inversion $\delta = w_c\sqrt{f_{\mathrm{gap}}/3}$ is within 0.5% of the exact root across the Gifford *calibration/design* band ($f_{\mathrm{gap}} \le 5.76\times10^{-4}$, device #33; worst case 0.47%), and within 0.9% across the wider *numerical sweep* band ($f_{\mathrm{gap}} = 3.63\times10^{-5}$ to 1.93×10^{-3}; worst case 0.86% at the top). Quote the band with the figure: the two differ by nearly a factor of two.

A.9.7 Cutoff law and prefactor decomposition (selectivity)

The cutoff law and its prefactor decomposition are numbered in Section A.4 (Eqns 1 and 2) and are restated here *unnumbered*, for dimensional analysis only, so that one equation carries one number:

$$d_{\mathrm{crit}} = \kappa_{\mathrm{app}}\,w_c\,\sqrt{f_{\mathrm{gap}}/3}, \qquad \kappa_{\mathrm{app}} = \kappa_{\mathrm{steric}}\,K_{\text{in-plane}}\,K_{\mathrm{depth}}\,\sqrt{c}\,C.$$

Dimensional analysis: κ_{app} is dimensionless (product of dimensionless factors), $[\sqrt{f_{\text{gap}}/3}] = 1$, so $[d_{\text{crit}}] = [w_c] = \text{m}$. ✓Each factor is a pure number (Table 4); $\kappa_{\text{steric}} = 2$, $K_{\text{in-plane}} = 1.000$, and the depth-folded residual $K_{\text{depth}}\sqrt{c}\,C = \kappa_{\text{app}}/\kappa_{\text{steric}} = 2.4569$ (depth-separated $\sqrt{c}\,C = \kappa_{\text{app}}/(\kappa_{\text{steric}}K^{\text{CFD}}_{\text{depth}}) =$ 2.7574/2.5999/2.5044 at $\alpha = 0.80/1.40/2.69$).

A.9.8 Depth-correction factor: analytic and CFD

$$K^{\text{th}}_{\text{depth}}(\alpha) = \sqrt{\frac{3}{c_2(\alpha)}} = \lim_{x\to 0}\frac{\delta_{3D}(\alpha)}{\delta_{2D}}, \qquad \frac{|K^{\text{CFD}}_{\text{depth}} - K^{\text{th}}_{\text{depth}}|}{K^{\text{CFD}}_{\text{depth}}} \le 0.6\% \tag{10}$$

Dimensional analysis: ratio of two widths, dimensionless. ✓$K^{\text{th}}_{\text{depth}}$ is evaluated from the rectangular-duct velocity field of aspect ratio $\alpha = h/w_{\text{ch}}$; $K^{\text{CFD}}_{\text{depth}}$ is the single-gap CFD measurement. **These are two distinct quantities and must not be averaged or interchanged.** Analytic: 0.8879/0.9500/0.9794. CFD: 0.891/0.945/0.981. Actual deviations 0.349%, 0.530%, 0.160%; the 0.6% figure is the *bound*, not the observed value. *The denominator is* $K^{\text{CFD}}_{\text{depth}}$, the measured factor, because the residual is defined against the measurement; dividing by $K^{\text{th}}_{\text{depth}}$ instead shifts the middle deviation from 0.530% to 0.527% and would no longer match the gated value. The analytic set is canonical for derivations; the CFD set is canonical for the depth-separated residual, which is defined against it.

A.9.9 Filtrate fraction: two identities, two roles

The filtrate fraction has two representations. They evaluate to the same realized F for a self-consistent device satisfying the model assumptions, but they expose different dependencies and should not be written as one undifferentiated chain of equalities.

Outlet divider (yield law).

$$F_{\text{outlet}} = \frac{2R_c}{R_s + 2R_c} \tag{11}$$

Dimensional analysis: numerator and denominator both $\text{Pa}\,\text{s}\,\text{m}^{-3}$, so F is dimensionless. ✓

Derivation under PE (P1.4) and $Q_s(0) = 0$ (P1.5). With outlet pressure equilibration $p_{\text{center}}(L) = p_{\text{side}}(L)$, the current-divider rule on the equivalent network (two side channels R_s in parallel, sharing the outlet node with the center channel R_c) gives $Q_s(\text{one side}) = R_c Q_{\text{in}}/(R_s + 2R_c)$, hence $F = 2Q_s/Q_{\text{in}} = 2R_c/(R_s + 2R_c)$. The hydraulic–electric analogy underpinning this is [10, 11].

Decoupling. f_{gap} does not appear in Eqn 11: F_{outlet} is fixed by outlet geometry (R_c, R_s) alone, while the cutoff is fixed locally by $\delta = w_c\sqrt{f_{\text{gap}}/3}$. *This absence is the decoupling.* The two axes are therefore separately addressable, and the orthogonality holds *because N is free*; pinning N re-couples them.

Cumulative march (internal extraction history).

$$F_{\text{recursion}} = 1 - \prod_{i=1}^{N}\left[1 - 2f_{\text{gap}}(i)\right] \tag{12}$$

Dimensional analysis: each factor dimensionless; product dimensionless. ✓Small-f_{gap} limit: $F \approx 1 - \exp(-2\sum_i f_{\text{gap}}(i))$; constant f_{gap}: $F \approx 1 - \exp(-2Nf_{\text{gap}})$. This form describes how cumulative extraction unfolds along the ladder; it does *not* exhibit the decoupling, because f_{gap} appears explicitly.

A.9.10 Hagen–Poiseuille / Ohm's-law analogue

$$\Delta P = R \cdot Q \tag{13}$$

Dimensional analysis: $[R \cdot Q] = \text{Pa}\,\text{s}\,\text{m}^{-3} \cdot \text{m}^3\,\text{s}^{-1} = \text{Pa} = [\Delta P]$. ✓The load-bearing identity of the hydraulic–electrical analogy ($P \leftrightarrow V$, $Q \leftrightarrow I$, $R \leftrightarrow R$): "pressure drop, flow rate, and flow resistance in microchannels are analogous to electrical voltage, current, and resistance, respectively" ([11, p. 3312]), and the circuit-analogy design paper states the same mapping as "correlations of pressure to voltage, volumetric flow rate to current, and hydraulic to electric resistance" ([10, p. 515]).

A.9.11 Reynolds number and flow regime

$$\mathrm{Re} = \frac{\rho U L_{\mathrm{char}}}{\mu} \tag{14}$$

Dimensional analysis: dimensionless (using $\mathrm{Pa\,s} = \mathrm{kg\,m^{-1}\,s^{-1}}$). ✓

CIF regime check (Dinh, water-like fluid). Dinh's channel depth is $h = 140$ µm and the middle channel narrows to $w_c \approx 52$ µm at the outlet, giving a hydraulic diameter $D_h = 2w_c h/(w_c + h) \approx 75.8$ µm. At $Q = 30\ \mathrm{\mu L\,min^{-1}}$ through the outlet cross-section (the whole flow through the narrowest section, a worst case that bounds the design code's local-flow value), $U = Q/A \approx 0.069\ \mathrm{m\,s^{-1}}$, so the *channel-scale* $\mathrm{Re} = \rho U D_h/\mu \approx 5.2$ (rising to ≈ 16 at $90\ \mathrm{\mu L\,min^{-1}}$). PRP is more viscous than water, so its Reynolds number is lower in proportion; none of the three papers states PRP's viscosity. The Reynolds number *at the gap* is obtained not by rescaling the length but by rescaling the *flux*: a gap carries only the diverted fraction f_{gap} of the channel flow, so $\mathrm{Re_{gap}} \approx f_{\mathrm{gap}}\,\mathrm{Re_{channel}} \approx 10^{-3} \times 5.2 \approx 5.2 \times 10^{-3} \ll 1$, evaluated at $f_{\mathrm{gap}} \approx 10^{-3}$, which is the *top* of the published lineage range rather than a margin above it. Gifford's 33-device *calibration* array caps lower, at 5.76×10^{-4} (device #33); his separate bead-separation devices were built with "a range of values of" f_{gap} "(2.2–10.0 $\times 10^{-4}$)" ([1, p. 4500]), reaching 1.0×10^{-3}, and no published lineage device exceeds it. The argument record's scope gate evaluates the same product. The design code's Stokes gate (M14) instead computes the gap's own Reynolds number, over the gap's $G \times h$ section at the first row's gap flow, which exceeds this product by $(w_c + h)/(G + h)$ (about 1.2 for Dinh's 19 µm gaps and outlet width), and a lamina-shear Re_δ, and in a cutoff design refuses either above 10^{-2}. **Do not confuse this with the wider band 3.63×10^{-5} to 1.93×10^{-3} quoted under "Skim cubic and dividing-streamline width" above**, which is the span of a deliberately widened numerical sweep used to demonstrate decoupling over a decade, not a range any built device occupies. *The margin is worth stating plainly*: at the top of that sweep, $f_{\mathrm{gap}} = 1.93 \times 10^{-3}$, the same scaling would give $\mathrm{Re_{gap}} \approx 0.010$ (the design code's own measure about a fifth higher, just past its gate), which is where the creeping-flow premise stops having room. The premise is comfortable for built devices and marginal at the top of the test sweep. *(A length-rescaling estimate gives different numbers and should not be conflated with this one: at the same U, using the* 19 µm *gap width gives* $\mathrm{Re} \approx 1.3$ *and using the outlet lamina width* $\delta \approx 0.31$ µm *gives* ≈ 0.02*. The flux argument is the one the scope gate uses.)* The steric-skimming junction is therefore in the creeping (Stokes) regime while the channel flow is low-Reynolds laminar. **The framework is therefore not globally Stokes**: it is scoped to rigid, approximately spherical particles in a Newtonian, nearly constant-viscosity fluid, with creeping-flow conditions *at the gap and lamina scales that govern particle selection.* Deformable particles, cell crowding and whole-blood rheology are outside the validated envelope. *(An earlier revision used* $L_{\mathrm{char}} = 30$ µm *and* $U = 0.22\ \mathrm{m\,s^{-1}}$ *and overstated the regime as "Stokes"; the steric derivation is unaffected because it rests on the gap-scale* $\mathrm{Re} \ll 1$*.)*

A.9.12 Entrance length and the quasi-developed assumption

$$\frac{L_e}{D_h} \simeq \left[0.619^{1.6} + (0.0567\,\mathrm{Re})^{1.6}\right]^{1/1.6} \xrightarrow{\mathrm{Re}\to 0} 0.619 \tag{15}$$

Dimensional analysis: ratio of two lengths, dimensionless. ✓

The kernels assume fully developed flow, and strictly it is not. The correct laminar entrance length has a *Stokes floor* and does not vanish as $\mathrm{Re} \to 0$; the naive $L_e = 0.05\,\mathrm{Re}\,D_h$ goes to zero in this regime and is wrong here. The correlation is Durst *et al.* [19] for developing *pipe* flow, applied here through the hydraulic diameter D_h; the 0.619 constant is that correlation's low-Reynolds limit. The rectangular-duct coefficient differs slightly, and [19] is the one citation in this section whose source is not held in the verification corpus, so the constant is taken as published. The conclusion below does not turn on it.

Evaluated on this document's own geometry, rather than quoted: at the Dinh inlet ($w_c = 75$ µm, $h = 140$ µm, $D_h = 97.7$ µm) the floor gives $L_e \approx 60.5$ µm; at the outlet ($w_c = 52$ µm, $D_h = 75.8$ µm) it gives $L_e \approx 46.9$ µm. Against the lineage row pitch $\Lambda = G + g_p = 46.1$ µm, which the record measures from Gifford's Fig. 4 ($G = 20.5$, $g_p = 25.6$), this is 1.31 at the inlet and 1.02 at the outlet. *Dinh publish no row pitch*: their Fig. 2 gives the gap width, the taper and the depth but no post width or spacing, so an estimated Dinh pitch near 49 µm (giving 1.23 and 0.96) is an inference, not a source figure, and the traceable Gifford value is quoted here instead. On either pitch the ratio exceeds unity at the inlet and sits at unity

at the outlet. **The entrance length is therefore comparable to the row pitch and exceeds it over much of the device: the flow does not classically redevelop between rungs.**

This is expected to be benign, for a stated reason: each rung perturbs the base flow by a relative amount of order f_{gap} (about 10^{-4}, and at most 10^{-3} across the published lineage range), so the error this makes in a segment's resistance is expected to be small. The design specification puts that error at $O(f_{\mathrm{gap}}^2) \sim 10^{-8}$; that order needs the first-order term to cancel, which this record does not show, so it is not used here as a bound. The conclusion does not turn on the exact value of L_e: it is the same whether L_e/Λ is 0.96 or 1.31. Report L_e/Λ; do not gate on it. Gate instead on the perturbation, which the design specification identifies with the design code's gap-dominance gate (M13).

A.9.13 Platelet volume and the Dinh threshold

$$V_{\mathrm{sph}} = \frac{4}{3}\pi a^3 \tag{16}$$

Dimensional analysis: $[V_{\mathrm{sph}}] = [a]^3 = \mathrm{m}^3$. ✓**Numerical (12 fL):** $a \approx 1.42$ µm, $D = 2a \approx 2.84$ µm (volume-equivalent sphere; a geometric conversion, not a measured diameter). Dinh report $2a \approx 4.8$ µm for the actual discoid 12 fL platelet ([5, p. 917]); see the volume-convention note in Section A.5.

A.9.14 Multi-variant ratio (Dinh V1, V3)

$$\mathcal{R} = \frac{\ln(1 - F_{V3})}{\ln(1 - F_{V1})} \tag{17}$$

Dimensional analysis: ln of dimensionless argument; ratio dimensionless. ✓**Origin.** With $F = 1 - \exp(-2N f_{\mathrm{gap}})$, $\ln(1-F) = -2N f_{\mathrm{gap}}$ and $\mathcal{R} = (N_{V3}\, f_{\mathrm{gap},V3})/(N_{V1}\, f_{\mathrm{gap},V1})$, sensitive only to the cumulative f_{gap}-product.

A.9.15 Unit-consistency summary

Each equation above was checked inline. The table gathers them so the whole set can be scanned at once, and records the convention each check runs under, because a dimensional identity that holds only under a convention is not established until the convention is named. Three rows come from Section A.4 and are checked here rather than there.

Table 8: Unit-consistency check for every load-bearing equation, both axes and the validation layer, with the convention each check assumes. Every entry matches; the column that carries information is the last.

Equation	LHS	RHS	Convention assumed
Eqn 4 (Gifford flow recursion)	$\mathrm{m}^3\,\mathrm{s}^{-1}$	$\mathrm{m}^3\,\mathrm{s}^{-1}$	f_{gap} carries the row index; Gifford's own form has none
Eqn 5 (Gifford continuity)	1	1	bilateral extraction, hence the factor 2
Eqn 6 (Yang series)	$\mathrm{Pa\,s\,m}^{-3}$	$\mathrm{Pa\,s\,m}^{-3}$	$h \le w_{\mathrm{ch}}$; $w_{\mathrm{ch}} \in \{w_c, w_s\}$, not the lamina w
Eqn 7 (Dinh f_{gap} definition)	$\mathrm{m}^3\,\mathrm{s}^{-1}$	$\mathrm{m}^3\,\mathrm{s}^{-1}$	per gap, per side
Eqn 8 (Xia corrected scaling)	1	1	the *corrected* form; the 2016 printing inverts the widths
Eqn 9 (skim cubic / δ)	1 / m	1 / m	two statements in one display: the cubic is dimensionless, δ is a length
Eqn 1 (cutoff law, d_{crit})	m	m	leading skim $w_c\sqrt{f_{\mathrm{gap}}/3}$, the form κ_g was fitted with
Eqn 2 (κ_{app} decomposition)	1	1	every factor a pure number
Eqn 3 (kernel–prefactor conversion)	1	1	*array-mean* $\overline{c_2}$ over the calibration widths, not a per-device c_2

Table 8 (continued)

Equation	LHS	RHS	Convention assumed
Eqn 10 (depth factor $K_{\text{depth}}^{\text{th}}$)	1	1	deviation denominator is $K_{\text{depth}}^{\text{CFD}}$, the measured factor
Eqn 11 (F_{outlet})	1	1	outlet pressure equilibration and $Q_s(0) = 0$
Eqn 12 ($F_{\text{recursion}}$)	1	1	cumulative over N rows; not the outlet identity
Eqn 13 (Hagen–Poiseuille analogue)	Pa	Pa	$P \leftrightarrow V$, $Q \leftrightarrow I$, $R \leftrightarrow R$
Eqn 14 (Reynolds number)	1	1	channel scale; the gap value rescales the *flux*, not the length
Eqn 15 (entrance length L_e/D_h)	1	1	the Re $\to$ 0 Stokes floor, 0.619
Eqn 16 (sphere volume)	m^3	m^3	volume-equivalent sphere, not a measured discoid diameter
Eqn 17 (multi-variant ratio)	1	1	constant-f_{gap} exponential limit

All seventeen are unit-consistent. *Dimensional consistency is necessary and not sufficient*, and the conventions in the last column are where the substantive assumptions live: the check confirms each equation is well formed under them, not that they hold.

A.10 Source-Quote Validation Index (verbatim)

Direct quotations anchoring each unit convention and adjudicated criterion, organized by source. Quotes are minimal; the page/figure/equation locator does the identifying work. A consolidated claim→source→anchor table follows (Table 9).

What needs a quotation, and what does not. Not every statement in this appendix should carry one, and forcing quotations onto mathematical or computed claims would weaken rather than strengthen them. The standard applied here is: *every externally attributable premise has a source anchor; every mathematical claim has a derivation; every computed claim has a reproducible computation; every measured claim has a traceable datum; every interpretive claim has an explicit evidentiary status.* Identities such as $D = 2a$, $D_h = 2wh/(w + h)$ and Eqn 11, the series inversion, the CFD values, the computed Reynolds numbers and the uncertainty ratios belong to the second, third and fourth categories and are marked as such rather than quoted.

How this index is maintained. Table 9 is *generated* from the quotation register, not typed. Every row is verified on every build: its text against the primary source and, where its page can be resolved, its page against the paginated source; rows marked † are text-verified only. Locators are corrected in the register and propagate here on the next build.

Gifford 2014 [1]

p. 4498 (f_{gap} regime): Gifford describes "devices that remove less than 0.05% of the fluid from the central channel at each gap," i.e. $f_{\text{gap}} < 5 \times 10^{-4}$. *This is the regime the source reports, not a design target it states.*
p. 4498 (Eqn 1): $Q_s(i) = Q_s(i-1) + f_{\text{gap}} Q_c(i-1)$ (no row index on f_{gap}).
p. 4498 (Eqn 3): continuity, from "the total volumetric flow within the device before and after each gap must be constant."
p. 4498 (Eqn 5): the rectangular-channel resistance carries citation [20] (= Yang/Ündar/Zahn, [6]); "L is the length, w is the width and d is the depth of the channel." Clarified on **p. 4499**: L is "the size (along the direction of flow) of an obstacle in the device."
p. 4500 (calibration): "33 test devices to explore a wide range of f_{gap} values ranging linearly from" 6.40×10^{-5} (device #1) to 5.76×10^{-4} (device #33); f_{gap}^{*} read visually; $f_{\text{gap}} = 3.34 \times 10^{-4}$ selected between beads of diameter 4.7 µm and 8.3 µm (Fig. 4 caption).

p. 4500, Fig. 3 caption (depth *unreported*): states only $w_c = 100, 125, 150$ µm, no depth (the 80 µm value belongs to the Fig. 4 device).
p. 4503: "the size cutoff is determined primarily by the width of the streamlines diverted into the side (filtrate) channels."

Yang 2006 [6] (resistance series)

p. 875: its Eqn 3 is "the analytical solution of flow resistance within a rectangular duct": the series Gifford's Eqn 5 cites and that Eqn 6 reproduces.

Xia 2016 [2] / 2022 Correction [3]

Correction p. 1 (verbatim): the article "contains an error in equation 1, where the terms $w_c(i)$ and $w_c(\text{ref})$ are inverted"; it should read $f_{\text{gap}}(i) = (w_c(\text{ref})/w_c(i))^2 \, f_{\text{gap}}(\text{ref})$.

Lezzar 2022 [4] (third lineage device)

p. 2: the CIF design in that study targets a 6 µm critical diameter, and "any CIF design must incorporate thousands of gaps." The unpublished gap count N is the quantity to which the per-gap over-extraction factor $\sqrt{c}$ is tied (Section A.4). *Verification note:* two of the three registered Lezzar quotes are the softest verifications in the register, one passing under ligature-tolerant matching and one against a page image rather than a text layer; they are not equivalent in strength to the exact-match anchors elsewhere in this index.

Dinh 2024 [5]

p. 913 (12 fL threshold): "PLTs with cell volumes larger than 12 fL are commonly considered 'large'," the P-LCR classification cut. *A classification threshold, not a cutoff.*
p. 915 (f_{gap} definition): a CIF device operates "by siphoning a relatively small (and precisely calculated) fraction, $f_{\text{gap}}(i)$, of the flow in the middle channel, $Q_m(i)$," so $Q_{\text{gap}}(i) = f_{\text{gap}}(i) Q_m(i)$.
p. 916: the taper, "progressively increasing the width of the side channels and decreasing the width of the middle channel."
p. 916, Fig. 1 caption (DESELECTED, $\kappa_{\text{steric}} = 1$): "about half the width of the extracted flow lamina", superseded by the main text and Inglis; the two Dinh readings are formally inconsistent, so one must be retracted.
p. 917, main text (ADOPTED for κ_{steric}): particles with a radius smaller than the lamina width ($a < w$) are carried into the side channels; larger particles ($a > w$) remain in the middle channel.
p. 917 (discoid platelet): "recovery of 12 fL PLTs ($2a \approx 4.8$ µm)", the physical discoid diameter (cf. 2.84 µm equivalent-sphere; see Section A.5).
p. 917, Fig. caption (geometry): gaps 19 µm-wide; middle channel narrows 75 → 52 µm; depth of all channels 140 µm.
p. 919, Fig. 4 caption (validation dispersion): "Values shown are mean ± standard deviation (panel A: $n = 3$; panel B: $n = 4$)." The same caption reports the filtrate volume recoveries, $50.5 \pm 3.3\%$ and $64.9 \pm 3.6\%$.

Yamada & Seki 2005 [12] (geometric exclusion)

p. 1233–1234: a particle centre cannot approach a sidewall closer than one particle radius, and the method uses that fact. This is the geometric-exclusion premise beneath κ_{steric}; it is *not* a source for the finite-particle factor C, which is cited to [15] and [17].

Davis 2008 [18] (cross-family DLD comparison)

pp. 14, 18, 22–24: the DLD critical diameter is about twice the width of the first streamline; the correlation with terms $A = 1.4$ and $B = 0.48$ is fitted to critical diameters measured on "about 20 different devices" and was used to design the thresholds the thesis reports, and its parabolic approximation is checked against

simulation. Used only in the cross-family comparison (Section A.11), never to fix a CIF constant. *Verification note:* the archived text carries the thesis's printed page numbers, and these locators are verified against them.

Inglis 2006 [13] (steric basis)

p. 656: the boundary-streamline fraction satisfies the parabolic-profile cubic $\varepsilon = 3y^2 - 2y^3$ (identical to the CIF skim cubic), and the critical diameter is twice the first-streamline width, which Inglis writes $D_c = 2\beta$ (its Eqn 1; Greek β is his symbol, rendered b here per the notation convention), giving $\kappa_{\text{steric}} = 2$.

Inglis & Herman 2013 [14] (lamina / branch flow)

p. 1724: a particle is "restricted away from walls by a distance equal to their radius," so a boundary stream of width b sets a cutoff radius b (equivalently $\kappa_{\text{steric}} = 2$); the abstract reports "high filtration ratios (14–29%) on beads and fungal spores" with "a sharp size cutoff."
p. 1727 (Methods): "The measured depth of the SU-8 mold was 48" µm, against 50 µm designed.
p. 1728, Table 1: design versus measured cutoff diameters ($7.2/5.0 = 1.44$; $8.9/6.0 = 1.48$; $9.7/8.0 = 1.21$).
p. 1729 (Discussion): "the experimentally observed cutoff diameters are significantly higher than the cutoff sizes determined from our design process," and the shallower channel "increases the main channel resistance more than it does the branch channel resistance" (see Section A.11).

MOPSA 2017 [15] (finite-particle method class)

p. 2: the simulation "treats the particle as a two-dimensional rigid circular object instead of a single point": the finite-body *method class* available for computing C, not an establishment of the factor itself.

Doyeux *et al.* 2011 [17] (factor-C mechanism)

p. 359 (abstract): near an asymmetric bifurcation the finite-sphere attraction is "directed towards the low flow rate branch" (anti-Zweifach–Fung): the *mechanism* that raises the measured cutoff above the steric value ($C > 1$), not a derivation of the multiplicative form. *Locator note:* the register cites *JFM* pagination while the archived artifact is the arXiv preprint, which paginates differently; the text verifies, the page does not.

Bruus 2008 [16] (resistance limit)

The rectangular-duct series reduces to $R = 28.4\,\mu L/h^4$ at the square cross-section $h = w$ (Eqn 6 limit check). This claim is *self-derived* rather than quoted: the registry's independent square-duct anchor is $12/B(1) = 28.4542$, checked by the derivation record rather than by the quotation gate.

Formal Argument

P1.4 (PE outlet condition): $p_{\text{center}}(L_{\text{dev}}) = p_{\text{side}}(L_{\text{dev}})$ at the outlet. *The physical condition is not introduced here.* The lineage source states that its resistance-ratio calculation holds "provided the hydrostatic pressures in the output collection areas are kept equal" ([1, p. 4500]). What is this work's is the *formalisation* of that condition as an equality of outlet-node pressures, and its use as the premise of the closed-form filtrate identity (Eqn 11).
P1.5 (inlet seal): $Q_s(0) = 0$; side channels accumulate filtrate only through the gap network. *This one is a numbered assumption of this work and carries no external anchor.*
Both are stated and used as premises in Section A.9.

Consolidated index. Table 9 maps every load-bearing claim to its source location and a short verbatim anchor, so a third party can confirm each in seconds. It is generated from the quotation register; do not edit it by hand.

Table 9: Claim → source (location) → verbatim anchor. Generated from the quotation register. Every row's *text* is verified against the primary source on every build; *pages* are checked where they can be resolved: against per-page images, or, for the Davis thesis, against the page numbers its text carries. Rows marked † are text-verified but not page-verified.

Claim in glossary	Source (location)	Verbatim anchor (short)
V&V-20 defines u_D to include all effects of averaging	[20, p. 1-6]	"includes all effects of averaging, includes all random and systematic uncertainty components"
Critical diameter is twice the first-streamline width (independent corroboration)	[18, p. 14]	"about twice the width of the first streamline"
DLD critical-diameter correlation, terms A and B	[18, p. 23]	"with terms A=1.4 and B=0.48"
That correlation was used to design the thresholds the thesis reports	[18, p. 24]	"equation 2-22 was used to design all the thresholds discussed herein"
Critical diameters measured across about twenty devices	[18, p. 22]	"about 20 different devices over a range of G from 1.3 micron to 38 micron"
Parabolic approximation checked against simulation	[18, p. 18]	"The Femlab simulation shows this simple parabolic approximation is a reasonable estimation for the velocity . . . "
Definition of the per-gap extraction fraction	[5, p. 915]	"siphoning a relatively small (and precisely calculated)"
V1 filtrate-volume recovery, the yield-axis experimental datum (panel B, n=4)	[5, p. 919]	"Volume recovery in filtrate was 50.5 ± 3.3% for f* gap = 1.04"
Dinh selects f*_gap as an initial value alongside the initial channel width	[5, p. 915]	"the initial values for the width of the middle channel and the filtration fraction"
Dinh reports three initial f*_gap values across three designs	[5, p. 917 (Fig. 2 caption)]	"for f* gap values of 1.04 × 10−4 , 1.12 × 10−4, and 1.28 × 10−4, respectively"
V3 filtrate-volume recovery, the yield-axis experimental datum (panel B, n=4)	[5, p. 919]	"64.9 ± 3.6% for f* gap = 1.28"
Dinh describes the taper as widening side channels and narrowing the middle	[5, p. 916]	"progressively increasing the width of the side channels and decreasing"
A platelet-size classification threshold (not the cutoff)	[5, p. 913]	"cell volumes larger than 12 fL are commonly considered"
Physical discoid diameter of a 12 fL platelet	[5, p. 917]	"recovery of 12 fL PLTs (2a ≈ 4.8 µm) in the filtrate"
Reported dispersion is standard deviation, with panel sample sizes	[5, p. 919]	"Values shown are mean ± standard deviation (panel A: n = 3; panel B: n = 4)"
Device operating flow rate	[5, p. 915]	"a CIF device operating at a flow rate of 30"
Smallest platelets are recovered in the filtrate below the fluid rate	[5, p. 917]	"recovery of the smallest 2 fL PLTs (a ≈ 1 µm) in the filtrate was consistently lower than the percent of extracted fluid volume"
Dinh's own explanation: crowding by retained platelets	[5, p. 917]	"crowding of the flow laminae near the filtration gaps by retained"
Crowding impedes the outflow of the smallest platelets	[5, p. 917]	"impeded the free outflow of the smallest PLTs"
Platelet diameter is the major/equatorial axis of an oblate spheroid	[5, p. 917]	"range in diameter (major/equatorial axis, 2a) from about 2.0"
DESELECTED ($\kappa_{\text{steric}} = 1$ reading), superseded by the main text and Inglis; the two Dinh readings are formally inconsistent, so one must be retracted: Caption reading places the cutoff at half the lamina width	[5, p. 916]	"Particles with an effective radius smaller than about half the width of the extracted flow . . . "
ADOPTED for $\kappa_{\text{steric}} = 2$: Main-text capture criterion, at a = lamina width	[5, p. 917]	"For particles with a radius smaller than the width of the extracted flow lamina (a . . . "
Device geometry: gap width, middle-channel taper, channel depth	[5, p. 917 (Fig. 2 caption)]	"All filtration gaps in the linear section were 19 µm-wide. The middle channel narrowed from . . . "
Platelets are discoid, not spherical	[5, p. 913]	"1 µm-thick oblate spheroids (discoids) ranging from 2 to 7 µm in diameter"
Discoid platelets tumble near the channel walls	[5, p. 921]	"the characteristic tumbling motion oblate spheroids undergo when flowing near the channel walls"
Bead bracket, lower end: a 1 um bead is recovered in the filtrate at the fluid rate	[5, Fig. S1(A)]†	"fluid volume recovery in filtrate was 52% and bead recovery in filtrate was 54%"
Bead bracket, upper end: a 4.7 um bead is retained against the fluid	[5, Fig. S1(B)]†	"fluid volume recovery in filtrate: 50%, bead recovery in filtrate: 1.8%"
Finite particle size shifts the effective separation line at a bifurcation	[17, p. 368]†	"might play a role, due to the finite size of the flowing objects"
Finite-sphere attraction is toward the low-flow branch (mechanism for C ¿ 1)	[17, abstract]†	"such an attraction does not occur but is, on the contrary, directed towards the low . . . "

continued on the next page

Table 9 (continued)

Claim in glossary	Source (location)	Verbatim anchor (short)
The apparent high-flow enrichment is an inlet-distribution effect	[17, abstract]†	"the enrichment in particles in the high flow rate branch is shown to be mainly . . . "
Flow recursion, Eqn 1	[1, p. 4498]	"Qs(i) = Qs(i − 1) + fgapQc(i − 1)"
Gifford built four devices spanning f_gap from 2.2e-4 to 10.0e-4	[1, p. 4500]	"four individual devices with a range of values of fgap (2.2–10.0 × 10−4"
Gifford names Eqn 5 variables: length, width, depth of the channel	[1, p. 4498]	"L is the length, w is the width and d is the depth of the . . . "
Gifford clarifies L as an obstacle size along the flow direction	[1, p. 4499]	"the size (along the direction of flow) of an obstacle in the device"
The Fig. 4 device is 80 um deep with a 100 um central channel	[1, p. 4500 (body; cf. the Fig. 4 caption, which gives 80 µm ± 3 µm)]	"The depth of the device shown in Fig. 4 was 80 µm; the width of the central channel was 100 µm"
Gifford describes the gaps as wide, about 20 um	[1, p. 4498]	"wide (20 µm) gaps"
Gifford names the total number of gap rows	[1, p. 4499]	"the necessary total number of rows of gaps"
Gifford defines f*_gap as the per-gap fraction below which beads are retained	[1, p. 4500]	"the filtration fraction per gap below which the beads of a spe"
Continuity, Eqn 3, from incompressibility	[1, p. 4498]	"the total volumetric flow within the device before and after each gap must be constant"
The reported per-gap extraction regime (not a stated design target)	[1, p. 4498]	"devices that remove less than 0.05% of the fluid from the central channel at each gap, i.e. fgap ¡ 5 × 10−4"
The 33-device calibration array and its f_gap span	[1, p. 4500]	"33 test devices to explore a wide range of fgap values ranging linearly from"
The f_gap selection bracket between two bead sizes	[1, p. 4500]	"retain beads with diameter 4.7 µm and diameter 8.3 µm"
Calibration figure gives three channel widths and no depth	[1, p. 4500]	"Effect of central channel width (wc = 100 µm, 125 µm, 150 µm)"
Centre- and side-channel flow rates defined	[1, p. 4498]	"a simple recursive eqn (1) for the volumetric flow rate in the central"
Centre and side segment resistances defined	[1, p. 4498]	"the resistance of a side channel segment, Rs(i), and the central channel segment resistance, Rc"
Outlet hydrostatic pressures are kept equal (the P1.4 condition)	[1, p. 4500]	"provided the hydrostatic pressures in the output collection areas are kept equal"
Cutoff is set by the diverted-lamina width, not the gap width	[1, p. 4503]	"the size cutoff is determined primarily by the width of the streamlines diverted into the . . . "
Platelets are approximately discoid	[1, p. 4502]	"approximately discoid"
Platelet-rich plasma behaviour in the lineage device	[1, p. 4502]	"we observed the behavior of PRP flowing through the same three arrays"
Pressure, flow and resistance map to voltage, current and resistance	[11, p. 3312]	"pressure drop, flow rate, and flow resistance in microchannels are analogous to electrical voltage, current, . . . "
Inglis derives the streamline width from a parabolic profile through the gap	[13, p. 656]	"By assuming a conventional parabolic flow profile through the gap, with zero velocity at the . . . "
Critical diameter is twice the first-streamline width, D_c = 2b	[13, p. 656]	"The critical particle diameter, Dc, for determining which path a particle will follow is then . . . "
First-streamline width defined by integrating the flow profile from the post edge	[13, p. 656]	"we integrate the flow profile"
First-streamline construction, second statement	[13, p. 655]	"is the width of the first streamline"
Branch-flow filtration ratios and a sharp size cutoff	[14, p. 1724]	"high filtration ratios (14–29%) on beads and fungal spores"
Measured cutoff diameters exceed the design values	[14, p. 1729]	"The experimentally observed cutoff diameters are significantly higher than the cutoff sizes determined from our design process. . . . "
Fabricated mould depth fell short of the design depth	[14, p. 1727]	"The measured depth of the SU-8 mold was 48"
A shallow mold raises main-channel resistance more than branch resistance	[14, p. 1729]	"increases the main channel resistance more than it does the branch channel resistance"
More branch flow raises the critical size at each branch	[14, p. 1729]	"more fluid flow in the branches, increasing the critical size at each branch"
Excluding a particle can need a streamline narrower than its radius	[14, p. 1729]	"excluding a particular size particle from a branch may require a streamline width that is as much as 50% smaller than the particle radius"
Inglis and Herman name a second contributing reason	[14, p. 1729]	"A second contributing reason is that the devices, as fabricated"
The finite-particle attraction runs contrary to the Zweifach-Fung effect	[14, p. 1729]	"contrary to the situation in the Zweifach-Fung effect"
The branches are narrow, so a shallow mould raises main resistance more	[14, p. 1729]	"because the branches are only 25"

continued on the next page

Table 9 (continued)

Claim in glossary	Source (location)	Verbatim anchor (short)
Same mechanism and exclusion premise, independent (non-lineage) group	[14, p. 1724]	"Assuming that particles do not affect streamlines and are geometrically restricted away from walls by . . . "
The same source's own caveat on that mechanism	[14, p. 1724]	"This is a frequent assumption in the microfluidic particle separation literature, that is not always . . . "
The same near-wall lamina integrated over the same rectangular-duct field	[14, p. 1724]	"the cutoff size is determined by the width of the boundary layer in the main . . . "
Third lineage device: the cutoff is set by the flow fraction, not the gap width	[4, p. 2 (Sci Rep PDF pagination)]	"It is the magnitude of this flow fraction—not the width of the gaps—that determines the . . . "
Third lineage device: 6 um critical-diameter target	[4, p. 2 (Sci Rep PDF pagination)]	"The critical diameter of the CIF design used in this study was 6"
Gap count is unpublished; sqrt(c) is tied to it	[4, p. 2 (Sci Rep PDF pagination)]	"any CIF design must incorporate thousands of gaps to achieve the desired"
Finite-body method class: the particle treated as a rigid disc, not a point	[15, p. 2]	"treats the particle as a two-dimensional rigid circular object instead of a single point"
Oh maps pressure to voltage, flow rate to current, and hydraulic to electric resistance	[10, p. 515]	"correlations of pressure to voltage, volumetric flow rate to current, and hydraulic to electric resistance"
Xia introduces the cutoff-diameter abbreviation c.d.	[2, p. 2 (Sci Rep PDF pagination)]	"particles above a certain size (or 'critical diameter', c.d.)"
RS-8 and RS-9 device parameters: cutoff, reference width and reference f_gap	[2, p. 6 (Sci Rep PDF pagination)]	"nominal c.d. of 8µm ("RS-8", wc(ref)= 100 µm and fgap(ref)= 7.2× 10−4) and 9 µm ("RS-9", wc(ref)= 100 µm and fgap(ref)= 8.6× 10−4"
OR-7 reference width and reference f_gap	[2, p. 10 (Sci Rep PDF pagination)]	"OR-7 was patterned using wc= 100µm and fgap= 5.8× 10−4"
The f_gap ramp exists to hold the cutoff constant along a tapering channel	[2, p. 9 (Sci Rep PDF pagination)]	"In order to maintain a consistent c.d. along the length of an RS-CIF device, the value of fgap can no longer be kept constant and must be scaled relative to"
The centre-channel taper exists to reduce shear, not to move the cutoff	[2, p. 4 (Sci Rep PDF pagination)]	"in order to reduce the maximum amount of shear stress experienced by particles within the . . . "
The 7 um cutoff is inherited by citation, not measured	[2, p. 10 (Sci Rep PDF pagination)]	"a combination of parameters we found corresponding to a c.d. of 7"
The citation points at the same figure the plateau is fitted from	[2, p. 9 (Sci Rep PDF pagination)]	"see Fig. 3 of Gifford et al."
Equation 1 of the 2016 paper is corrected: the two widths were inverted	[3, p. 1]	"This Article contains an error in equation 1, where the terms wc(i) and wc(ref) are inverted"
A particle centre is excluded from the wall by one radius	[12, p. 1234]	"the center position of a particle cannot be present within a certain distance from the . . . "
Geometric-exclusion premise, context sentence	[12, p. 1233]	"particles are concentrated and aligned onto the sidewalls"
The rectangular-duct resistance series the lineage resistance law cites	[6, p. 875]	"analytical solution of flow resistance within a rectangular duct"

[†]Text verified against the primary source; *page* not machine-verified (a figure anchor rather than a page, or preprint-versus-journal pagination).

Self-derived (not from a single source). The skim-cubic inversion $\delta = w_c[\sqrt{f_{\rm gap}/3} + \frac{1}{9} f_{\rm gap} + \cdots]$ is obtained by series-reverting $f_{\rm gap} = 3x^2 - 2x^3$ (verified term-by-term). The worked V&V-20 distances follow from the tabulated F-values with n=4. The Reynolds figures ($D_h \approx 75.8$ µm, $U \approx 0.069\ \mathrm{m\,s^{-1}}$, Re ≈ 5.2) follow from the Dinh geometry by $D_h = 2w_c h/(w_c + h)$ and $U = Q/A$. The square-duct limit $12/B(1) = 28.4542$ is a registry anchor. The kernel-pairing table is generated from the registry. All were re-derived independently.

A.11 Cross-Lineage Validation (Davis DLD; Inglis & Herman branch flow)

The cutoff prefactor's open residual predicts that a measured cutoff diameter sits *above* the idealized steric value $d_{\rm crit}^{\rm ideal} = 2\delta = 2w_c\sqrt{f_{\rm gap}/3}$. This is corroborated across three device families: CIF (Shevkoplyas lineage), deterministic lateral displacement (Davis), and branch-flow filtration (Inglis & Herman). *Within the CIF lineage itself there is no independence*: Gifford, Xia, Dinh and Lezzar share a laboratory, and it is the two non-CIF families that carry the out-of-lineage check.

What agrees, and what does not. The consilience is in the *sign* and the *floor*, not the magnitude. Every family's measured cutoff *diameter* exceeds the steric floor $\kappa_{\rm steric} = 2$ where the three are compared: on CIF's plateau, and across the whole reported range of the other two. (Below CIF's knee the apparent prefactor falls under the floor, to the sub-steric foot of Section A.4; that regime is the knee's own, and is not a counterexample.) **The magnitudes do not agree, and that is a result rather than a discrepancy to be harmonised away**: CIF's excess is roughly *twice* the DLD and branch-flow families'. Correctly normalised, DLD shows a smooth, knee-free apparent prefactor across $\varepsilon = 0.01$–0.5, bracketed by $\kappa_{\rm app} = 2.040$–2.659. *Those two bounds are two computations of the same DLD prefactor, not two laboratories, and they run in opposite senses*: Davis's eqn 2-22 falls with ε, from 2.659 to 2.459, while the exact Inglis *cubic* ([13], the skim construction of Section A.9) rises, from 2.040 to 2.449. At the top of the range the two agree to 0.4%, so the width of the bracket is a between-method gap at small ε, not size-dependence within either. Each is monotone and neither shows a knee. Read "Inglis" there as the cubic, not as Inglis & Herman, whose branch-flow data are the separate third family above. Within the examined DLD comparison datasets the normalized prefactor therefore does not exhibit the CIF knee: the size-dependence is CIF-specific

Table 10: Inglis & Herman 2013, Table 1: design vs. measured cutoff diameter and filtration ratio (FR). The measured cutoff exceeds the design in every case; the mean ratio is ≈ 1.4. ± values are CV over three runs.

Design D_c (µm)	Design FR	Measured FR	Measured D_c (µm)	Measured/Design
5.0	13.3%	$14.5 \pm 0.2\%$	7.2	1.44
6.0	18.3%	$20.4 \pm 0.5\%$	8.9	1.48
8.0	28.4%	$28.5 \pm 0.4\%$	9.7	1.21
Mean measured/design ratio				≈ 1.4

Table 11: Three-family excess over the steric floor $\kappa_{\text{steric}} = 2$, all expressed as ratios to that floor so they compare directly. The magnitudes differ, and that difference is a result.

Family (source)	Excess over the steric floor	Basis
CIF (Shevkoplyas)	$\kappa_{\text{app}}/\kappa_{\text{steric}} = 2.4569$	Gifford Fig. 3 plateau fit
DLD (Davis)	$\kappa_{\text{app}}/2 = 1.23$–$1.33$	recomputed from Davis eqn 2-22 [18]
Branch flow (Inglis & Herman)	measured/design = 1.21–1.48	[14], Table 1

within the validation corpus, not demonstrably a universal feature of streamline-skimming devices. *The factor-of-two difference is an observed quantitative distinction between CIF and the comparison families, not an identified physical cause. Note the two bases*: Table 11's DLD excess 1.23–1.33 is the Davis design correlation alone, which is the leg the "roughly twice" comparison rides on, whereas the wider bracket above adds the theoretical cubic and is used only to establish the absence of a knee. Halving the bracket would give a different and incomparable range. **Compare the depth-*folded* 2.4569 with the other two families, not the depth-separated triplet**: the published measured/design ratios are depth-inclusive.

Interpretation of the branch-flow leg. The authors state that the "experimentally observed cutoff diameters are significantly higher than the cutoff sizes determined from our design process" ([14, p. 1729]), and they themselves name two contributing reasons for it ("A second contributing reason is that the devices, as fabricated…", [14, p. 1729]), which correspond to the two non-unity factors of Eqn 2. *The correspondence is to mechanism classes, not to this work's specific factorization*, and the published data fix the *product* $\sqrt{c}\,C$ without fixing the split between its two factors. That degeneracy is a property of the available measurements, not a gap in the analysis: no published dataset varies c and C independently.

1. **Finite-particle displacement (C).** A finite sphere near a bifurcation is drawn toward the *low-flow-rate* branch, "contrary to the situation in the Zweifach-Fung effect" (Doyeux *et al.*, [17]; the contrast is Inglis & Herman's own phrase, [14, p. 1729]). They invoke the mechanism themselves, agreeing with Doyeux that "excluding a particular size particle from a branch may require a streamline width that is as much as 50% smaller than the particle radius" ([14, p. 1729]): the streamline that just excludes the particle is narrower than its radius, raising the measured cutoff.
2. **Over-extraction from increased branch flow ($\sqrt{c}$).** The realized SU-8 mold depth was 48 µm against a 50 µm design ([14, p. 1727]). Inglis & Herman attribute the resulting cutoff shift not to a skim-width change but to resistance: the shallower channel "increases the main channel resistance more than it does the branch channel resistance" "because the branches are only 25" µm wide, which will mean "more fluid flow in the branches, increasing the critical size at each branch" ([14, p. 1729]). *They give no magnitude for that shift*: the 48-versus-50 µm depth and the 25 µm branch width are geometry, not a quantification of the cutoff change, which is why the published data fix the product and not the split. This is per-gap *over-extraction*, the $\sqrt{c}$ factor, not the finite-aspect-ratio skim-width factor K_{depth}. *Note that this makes the* 1.21–1.48 *range a mixed measurement*: part of their excess is a fabrication deviation, so it is not a pure finite-particle figure.

The independent branch-flow literature therefore corroborates the sign and the plausible physical classes of the CIF residual; it does not identify the CIF-specific numerical factors c and C separately.

A.12 Master Cheat-Sheet

A single consolidated lookup table for both performance axes, the cutoff prefactor decomposition, the unification campaign, and the validation layer. Values in **bold** are fixed constants; w_c-dependent and device-dependent quantities are given as relations. *This table summarises; where it and a subsection differ, the subsection governs.*

Table 12: CIF master cheat-sheet. "Unit" is the practical CIF-lineage convention; all derivations use SI.

Symbol	Value / defining relation	Unit	One-line meaning
Geometry			
w_c	design input	µm	Center-channel width
$w,\ \delta$	$\delta = w_c\sqrt{f_{\text{gap}}/3}$ (theory); w empirical	µm	Diverted-lamina width ($b \equiv \delta \approx w$)

Table 12 (continued)

Symbol	Value / defining relation	Unit	One-line meaning
h	design input; Eqn 6 is written with $h \le w_{\mathrm{ch}}$, the two exchanged when $h > w_{\mathrm{ch}}$	µm	Channel depth
G, g_p, Λ	design inputs; $\Lambda = G + g_p$. *Lower-case g is the cubic kernel, not a length* (Table 1)	µm	Link width, post width, row pitch
ϕ	$\phi = G/\Lambda$ (computed); cf. ϕ_{cal} (external)	—	Wall porosity; one of three residual model gaps
N	design input	—	Number of gap rows
Yield axis			
f_{gap}	$Q_{\mathrm{gap}} = f_{\mathrm{gap}} Q_c$; Gifford's stated practical regime is $f_{\mathrm{gap}} < 5 \times 10^{-4}$	—	Per-side, per-gap extraction fraction
R_c, R_s	Eqn 6 (Yang series, [6])	$\mathrm{Pa\,s\,m^{-3}}$	Segment hydraulic resistances
F_{outlet}	$= 2R_c/(R_s + 2R_c)$, with no f_{gap}; this absence *is* the decoupling	—	Outlet divider under PE (P1.4) and $Q_s(0) = 0$ (P1.5)
$F_{\mathrm{recursion}}$	$= 1 - \prod_i (1 - 2f_{\mathrm{gap}}(i))$	—	Cumulative march; equals F_{outlet} for a self-consistent device
Π	$\prod_i (1 - 2f_{\mathrm{gap}}(i))$	—	Retained fraction
Selectivity axis (cutoff)			
d_{crit}	$d_{\mathrm{crit}} = \kappa_{\mathrm{app}} w_c \sqrt{f_{\mathrm{gap}}/3}$	µm	Critical (cutoff) diameter
$\widetilde{d}$	$\widetilde{d} = d_{\mathrm{crit}}/w_c$; plateau ≥ 0.035, foot < 0.012	—	Reduced size ratio (sets the bins)
x	root of $f_{\mathrm{gap}} = 3x^2 - 2x^3$; $x = \delta/w_c$	—	Reduced skim width (Inglis cubic)
β, γ	$\partial \ln d_{\mathrm{crit}}/\partial \ln f_{\mathrm{gap}} = \frac{1}{2}$ (≈ 0.51); $\partial \ln d_{\mathrm{crit}}/\partial \ln w_c = 1$ (≈ 1.02)	—	Cutoff scaling exponents ($\neq$ lamina width b)
$\kappa_{\mathrm{app}} \equiv \kappa_g$	supplied **4.9138** (95% CI [4.75, 5.08], CV 4.0%); refit 4.9137	—	Apparent prefactor, **cubic branch**
$\kappa_{F_{\mathrm{lam}}}$	**5.798** $= \kappa_g \sqrt{\overline{c_2}/3}$ at assumed $h_{\mathrm{cal}} = 80$ µm. **Not interchangeable with** κ_g: crossing shifts d_{crit} by 18.0% / −15.3%; the kernel choice alone shifts it by 14.1% to 15.6%	—	Prefactor, **exact** F_{lam} **branch**
κ_{steric}	**2** ($D_c = 2b$, Inglis's 2β; Dinh $a < w$)	—	Steric factor (radius = lamina)
$K_{\text{in-plane}}$	**1.000** (single-gap CFD)	—	In-plane funnelling (none)
$K^{\mathrm{th}}_{\mathrm{depth}}$	analytic 0.8879/0.9500/0.9794 at $\alpha = 0.80/1.40/2.69$ †	—	Depth factor, theorem
$K^{\mathrm{CFD}}_{\mathrm{depth}}$	CFD 0.891/0.945/0.981; $\lvert K^{\mathrm{CFD}}_{\mathrm{depth}} - K^{\mathrm{th}}_{\mathrm{depth}} \rvert / K^{\mathrm{CFD}}_{\mathrm{depth}} \le 0.6\%$, at most 0.531%. *The denominator is the CFD anchor*, matching the gated value	—	Depth factor, CFD
$\sqrt{c}\,C$	$K_{\mathrm{depth}} \sqrt{c}\,C = \kappa_{\mathrm{app}}/\kappa_{\mathrm{steric}} =$ **2.4569** (depth-folded); $\sqrt{c}\,C = 2.7574/2.5999/2.5044$ (depth-separated). Two factors; *three* model gaps	—	Open calibrated residual
α_{cal}	$(0.80, 0.64, 0.5333)$ at assumed $h_{\mathrm{cal}} = 80$ µm; $\Delta_{\mathrm{transfer}} = 20.3\%$. The +26.9% kernel/depth sensitivity is *ungated*, from a specification sweep this record does not reproduce	—	Calibration array; **unreported depth**
Particle			
a, D	$D = 2a$; $V_{\mathrm{sph}} = \frac{4}{3}\pi a^3$	µm	Radius / diameter (rigid sphere)
V_{12}	12 fL classification threshold → $D \approx 2.84$ µm equivalent sphere; $2a \approx 4.8$ µm (Dinh discoid)	fL	**Not** a minimum retained volume
Regime and envelope			
Re	channel ≈ 5.2; gap/lamina $\approx 5.2 \times 10^{-3} \ll 1$	—	Not globally Stokes; creeping at the selecting scale
L_e/Λ	$L_e \approx 46.9$–60.5 µm (outlet to inlet) against the Gifford Fig. 4 pitch $\Lambda = G + g_p = 46.1$ µm, i.e. $L_e/\Lambda \approx 1.02$–1.31	—	Exceeds the row pitch; expected benign either way (per-rung perturbation of order f_{gap})
u_{lo}, σ_0, ϕ_{cal}	supplied	—	External inputs, undischargeable
η	$\eta = 0.903$, a computed magnitude, not a free parameter (Table 7)	—	Realized-yield factor, MEASURED
Validation (V&V-20)			
E	$E = S - D$ (sim − data)	—	Validation comparison error; S and D are reused letters
α_{3D}	$(R_c/R_s)_{\mathrm{app}}/(R_c/R_s)_{\mathrm{Yang}}$; 0.811/0.660	—	Outlet correction reconciling 1-D with 3-D
u_D	the published SD, a *conservative* choice. V&V 20's u_D for a mean of n has random component SEM = SD/$\sqrt{n}$, reported alongside	—	Data uncertainty
u_{val}	$\sqrt{u^2_{\mathrm{num}} + u^2_{\mathrm{input}} + u^2_D}$	—	Combined validation uncertainty
F check	1-D closed form $+1.64\sigma$ / $+1.94\sigma$; 3-D COMSOL $+0.06\sigma$ / -0.58σ on the SD scale	—	The 3-D solve lands within 1σ on the SD scale, 2σ on SEM

Table 12 (continued)

Symbol	Value / defining relation	Unit	One-line meaning
Cutoff check	Dinh band $[1.25, 2.17]$ inside measured $(1.0, 4.7)$ µm; Xia $\kappa_{\text{app}} \approx 5.17$ vs 4.91, reported as a *lead*	µm	Both consistency checks: Dinh held out but weak, Xia not independent

† The κ_{app} fit uses the Gifford Fig. 3 array, whose depth is *unreported*; $\alpha = 0.8$ assumes the 80 µm depth of the *separate* Fig. 4 device, of which Gifford writes "The depth of the device shown in Fig. 4 was 80 µm; the width of the central channel was 100 µm" ([1, p. 4500]), and $K^{\text{th}}_{\text{depth}}$ varies across the three calibration widths (0.888/0.849/0.811 at $w_c = 100/125/150$ µm, i.e. $\alpha = 0.80/0.64/0.5333$). See Section A.4.

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